Contents

1 Introduction

2 Literature review

3 Methodology

4 Data

5 Analysis

6 Results

7 Conclusion

Reference

Appendix

An Introduction to Mental Health

What is Mental Health?

A person's emotional, psychological, and social well-being are all referred to as their mental health. It includes a person's thoughts, feelings, behaviours, and ways that they interact with people, manage stress, and make decisions. A person's capacity to function well in daily life and overall quality of life are both influenced by their mental health.

The key aspects of mental health include a person's emotional wellbeing which is the ability to convey one's emotions without being overwhelmed. Psychological wellbeing which includes having a positive self-image and the ability to manage stress and anxiety levels. Being able to have healthy and strong social connections is also a part of having good mental health. Apart from these, the ability to stay strong and resilient through difficult situations as well as seeking help at needed times are also part of mental Health.

Mental health is a spectrum, ranging from flourishing mental well-being to struggling with mental health disorders.

Mental Health Disorders

The spectrum of Mental Health that is struggling is our main topic for this project. What is Mental Health Disorders?

Mental health diseases, sometimes called mental illnesses or psychiatric disorders, are illnesses that have an impact on a person's thoughts, feelings, behaviours, and general psychological well-being. These illnesses can impair a person's capacity to carry out daily activities, uphold relationships, and deal with obstacles encountered on a regular basis. Mild to severe mental health illnesses can affect a person's quality of life significantly and can range in severity.

There are numerous varieties of mental health diseases, each with a unique set of signs, root causes, and therapeutic strategies.

Depression: characterised by enduring melancholy, pessimism, and a lack of enthusiasm for or enjoyment from activities.

Anxiety: involves overly worrying and fretting over several parts of existence. Rapid heartbeat and shortness of breath are common physical symptoms of sudden, severe panic attacks. high anxiety and social interactions are avoided.

Bipolar Disorder: Involves mood swings between periods of mania (elevated mood, increased energy) and depression.

Schizophrenia: A severe disorder that affects a person's thinking, emotions, and behaviour. It can involve hallucinations, delusions, disorganized thinking, and impaired social functioning.

Eating Disorders: Involves severe restriction of food intake and an intense fear of gaining weight. Characterized by episodes of binge eating followed by behaviours to compensate for overeating, such as vomiting or excessive exercise. Involves consuming large amounts of food in a short period of time without compensatory behaviours.

Substance Use Disorders: Occur when the use of substances (alcohol, drugs) leads to significant impairment in daily functioning, health, or relationships.

Combinations of genetic, biochemical, psychological, and environmental factors can lead to mental health issues. A person's quality of life can be greatly improved by early intervention and appropriate treatment, which may involve therapy, medication, and lifestyle modifications.

It's important to note that mental health is not a static state; it can change over time due to various factors such as life events, genetics, environment, and personal experiences. Taking steps to maintain and improve mental health is a proactive approach to overall well-being.

Mental health issues are a global concern that affects people of all ages, backgrounds, and cultures. They can have a significant impact on individuals, families, communities, and societies as a whole.

Mental health issues are widespread and affect millions of people worldwide. According to the World Health Organization (WHO), approximately 1 in 4 people will experience a mental health issue at some point in their lives. A wide range of diseases, from common ones like depression and anxiety to more serious ones like schizophrenia, are included in the category of mental health difficulties. Additionally, problems with substance abuse and addiction add to the load on the world's mental health. The stigma and discrimination associated with mental illness continue to be major obstacles to getting the right care and assistance. Isolation and recovery difficulties might result from stigma. International agencies like the World Health Organisation (WHO) are aiming to make mental health a priority on the international agenda, formulate regulations, and give funding to promote mental health care globally.

Literature Review

Focused topics for the Project.

As the project follows Mental Health issues and a study about it in a global level these where some of the questions that where focused to find answers for.

- Global Burden of Mental Illness: Analyze the dataset to determine the global burden of mental illness by examining the prevalence rates across different countries. Investigate variations in prevalence rates and identify factors contributing to differences in mental health outcomes globally.
- Longitudinal Analysis: Conduct a longitudinal analysis of the dataset to examine changes in the prevalence of mental illness over time in different countries. Identify temporal trends, explore potential factors driving these changes, and assess the effectiveness of mental health interventions and policies.
- Socioeconomic Factors and Mental Health: Investigate the relationship between socioeconomic factors (such as GDP, education, income inequality) and the prevalence of mental illness in different countries.
- Analyze the prevalence rates of eating disorders in various countries using the provided mental health dataset

In order to understand the cause of Mental illness in different regions it was necessary to read different research papers as well as literatures about people struggling with different mental Health

issues in different sectors. Which opened a pathway to understanding that different situations can end up in causing the same Mental Health Issues. The analysis write ups written by various researchers and professors mainly points toward that most mental illness is caused from educational and work sectors. And it shows that Peer pressure is one of the major reasons for Most Mental Health issues.

Bar Graph Plots

A bar graph, also known as a bar chart, is a graphical representation that displays data using rectangular bars. Each bar corresponds to a category or group, and the length or height of the bar represents the value or quantity associated with that category. Bar graphs are commonly used to compare the values of different categories or to show changes in data over time. A bar graph, also known as a bar chart, is a visual representation that uses rectangular bars to display and compare data across different categories or groups. Each bar corresponds to a category, and the length or height of the bar represents the quantity, frequency, or value associated with that category. Bar graphs are particularly effective in illustrating discrete data and showing relationships, trends, or comparisons.

In a bar graph, the x-axis typically represents the categories or groups being compared, while the y-axis represents the values or frequencies. The bars can be oriented vertically or horizontally, depending on the preference and the data's nature.

Bar graphs provide an intuitive way to present information, making it easy to identify patterns, disparities, or trends among different categories. They are widely used in various fields, including statistics, business, education, and research, to visualize data in a clear and accessible manner. Bar graphs are customizable, allowing users to add labels, titles, and color-coded bars to enhance the visual representation and convey insights effectively. Overall, bar graphs offer a straightforward and impactful method for displaying categorical data and facilitating data-driven decision-making.

In this project the bar graph is used to measure the impact of each mental health throughout that continent from the oldest data to the latest data available. This Gives an overview of how much each Mental illness impact the world throughout the world. Even though it does not give an accurate measurement, it can be used to get a rough idea of how each mental illness is having an impact globally which is important at the beginning of the research.

Trend Line Graph

A trendline graph, also known as a line graph, displays data points on a Cartesian plane and connects them with straight lines to show trends and patterns over time or across variables. Trendline graphs are commonly used to visualize how data changes over continuous intervals and to identify trends, fluctuations, or relationships between variables. A trendline graph, also known as a line graph, is a visual representation that displays data points as connected line segments. It is used to illustrate trends, patterns, and changes in data over a continuous range, such as time or a variable. In a trendline graph, each data point is represented by a marker, and a line is drawn to connect these markers, emphasizing the overall trend in the data.

Trendline graphs are particularly effective for depicting changes and relationships between two variables. The x-axis typically represents the independent variable (e.g., time) and the y-axis

represents the dependent variable (e.g., values or measurements). The line connecting the data points provides a visual representation of how the dependent variable changes in response to changes in the independent variable.

Trendline graphs are commonly used in various domains, such as economics, science, and social sciences, to visualize data trends, forecast future values, and identify patterns. They offer a clear and concise way to communicate complex relationships and variations in data, making them a valuable tool for data analysis and presentation. Customization options, such as adding labels and titles, allow users to enhance the graph's clarity and convey insights effectively.

The graph is used to identify the trend of the Mental illness throughout the years and also to try and find an initial relationship between any of the variables. It should also be noted that even though two variables may show a similar and closer trend, they will not always be a relation between them. The trend line can only give an overview, in order to have a better conclusion further research and study is always needed.

Correlation Graph

A correlation graph, also known as a correlation matrix or heatmap, is a visual representation of the relationships between multiple variables in a dataset. It uses color-coded cells to indicate the strength and direction of correlations between pairs of variables. Correlation graphs are commonly used to understand the patterns of association between variables and to identify potential relationships that might exist. A correlation graph, often referred to as a correlation matrix or heatmap, is a visual representation that displays the relationships and strengths of associations between multiple variables in a dataset. It is a valuable tool in statistics and data analysis for understanding how variables co-vary, providing insights into potential patterns and dependencies.

In a correlation graph, variables are represented by both rows and columns, forming a symmetric matrix. Each cell in the matrix contains a correlation coefficient, which quantifies the strength and direction of the relationship between two variables. Positive values indicate a positive correlation (both variables increase or decrease together), negative values indicate a negative correlation (one variable increases while the other decreases), and values close to zero suggest a weak or no correlation.

Correlation graphs are often visualized as heatmaps, where colours represent the magnitude of correlation coefficients. Darker or warmer colours indicate stronger correlations, while lighter or cooler colours suggest weaker or no correlations.

Correlation graphs are useful for identifying which variables are related and can assist in feature selection, multicollinearity detection, and understanding data relationships. However, they do not imply causation. Correlation graphs help researchers and analysts make informed decisions based on data patterns, aiding in predictive modelling and informed decision-making across various fields.

This graph gives a better grasp about what the relation between each variable are which helps in forwarding the study more accurately. This graph gives mostly accurate relationship between each variable, so the study can focus on relationship with stronger correlation.

Geographical Plots

Geographical plots, also known as maps, are visual representations of data on a geographical map to show spatial relationships, distributions, and patterns. Geographical plots, also known as maps, are visual representations of data and information on a geographic layout. They provide a spatial context for understanding patterns, distributions, and relationships across geographical regions. Geographical plots utilize cartographic elements like coordinates, shapes, and colours to convey information effectively.

Geographical plots are essential for fields such as geography, environmental science, epidemiology, and business analysis. They offer insights into various phenomena, such as population density, disease outbreaks, climate variations, and market trends.

Modern tools like geographic information systems (GIS) and data visualization libraries in Python (e.g., Folium, Geopandas) enable the creation of interactive and static geographical plots. These plots can display points, lines, polygons, and choropleth maps, allowing users to explore and analyse data on different spatial scales.

By visualizing data geographically, decision-makers can make informed choices, researchers can identify spatial patterns, and the public can better understand complex geographic relationships. Geographical plots enhance communication and provide a visual context that goes beyond tabular data, enabling insights that might not be apparent through traditional numerical analysis alone.

These plots helped in analysing the impact of each mental heath issue in each country. The darker the shade the higher the impact. This helped in understanding the countries to be focused on for each issue. These plots also helped in analysing the countries with highest Mental Health Issue Problems. Pictorial representations give a clear understanding about most issues. An analysis can occur by simply looking and understanding the details which helps in further analysis.

Hybrid models

Hybrid models, often referred to as hybrid algorithms or methods, combine the strengths of different techniques or approaches to enhance overall performance and achieve better results in various fields such as machine learning, optimization, and data analysis. These models leverage the complementary nature of multiple methodologies to overcome limitations and provide more accurate and robust solutions.

In hybrid models, different components or sub-models are integrated to work together harmoniously, exploiting their individual advantages while compensating for each other's weaknesses. This synergy often leads to improved predictive accuracy, better generalization, and increased adaptability to complex scenarios.

For instance, in recommendation systems, a hybrid model might combine collaborative filtering and content-based methods to offer more personalized and accurate suggestions to users. In finance, a hybrid optimization algorithm could blend gradient-based techniques with evolutionary strategies to optimize investment portfolios efficiently.

Hybrid models require careful design, as the integration of diverse techniques demands compatibility and synchronization. While they can be more complex to develop and fine-tune, their potential to outperform single-method models in challenging and diverse scenarios makes them a powerful tool for solving real-world problems.

Methodology

This project follows the Scientific Methodology which is a systematic approach done by researchers and scientists to investigate and understand natural phenomena through observation, experimentation, and logical reasoning. It involves a structured process aimed at formulating hypotheses, conducting controlled experiments, collecting empirical data, analysing results, and drawing evidence-based conclusions.

It begins by observing the changes caused by the topic of interest around the world and questioning the different probabilities. A hypothesis is a testable statement or educated guess that proposes a potential explanation for the observed phenomena. It serves as the basis for further investigation. Variables are manipulated and measured to observe how they interact and influence outcomes. During experiments, researchers collect quantitative and qualitative data to objectively document observations. Rigorous data collection methods ensure accuracy and reliability. Data is analysed using statistical methods and tools to identify patterns, trends, correlations, and other relevant insights. Based on the analysis conclusions are drawn about the credibility of the hypothesis. The scientific methodology fosters objectivity, repeatability, and the advancement of knowledge. It encourages an iterative process of refinement and revision, leading to a deeper understanding of natural processes. While the scientific method is a robust framework, it's important to note that not all inquiries can be addressed using this methodology, particularly in areas that require ethical or historical analysis.

Research Design: The project is focused on analysing the Mental Health issues around the world by producing different analysis methods and plots different questions are going to be answered. The questions that rise while analysing the given data is either answered using experiment results or by analysing research papers that are related to the project study.

Some of the objectives in concern in this project are

- Global Burden of Mental Illness: Analyze the dataset to determine the global burden of mental illness by examining the prevalence rates across different countries. Investigate variations in prevalence rates and identify factors contributing to differences in mental health outcomes globally.
- Longitudinal Analysis: Conduct a longitudinal analysis of the dataset to examine changes in the prevalence of mental illness over time in different countries. Identify temporal trends, explore potential factors driving these changes, and assess the effectiveness of mental health interventions and policies.
- Socioeconomic Factors and Mental Health: Investigate the relationship between socioeconomic factors (such as GDP, education, income inequality) and the prevalence of mental illness in different countries.
- Analyze the prevalence rates of eating disorders in various countries using the provided mental health dataset.

Data Analysis: in this project three different analytical methods are tried to use

Statistical Analysis: Statistical analysis is a systematic process of collecting, organizing, interpreting, and drawing conclusions from data. It involves using mathematical and statistical techniques to

uncover patterns, relationships, and insights that inform decision-making, scientific research, and problem-solving. The main steps in statistical analysis include data preprocessing, descriptive statistics, inferential statistics, and interpretation.

Descriptive statistics summarize and present data using measures like mean, median, and standard deviation, providing a clear overview of the dataset's central tendencies and dispersion. Inferential statistics, on the other hand, draw conclusions about populations based on sample data. Hypothesis testing, confidence intervals, and regression analysis are common inferential techniques that help assess relationships and make predictions.

Statistical analysis plays a crucial role across various disciplines, from business and economics to medicine and social sciences. It aids in identifying trends, making predictions, testing hypotheses, and validating theories. Properly conducted statistical analysis provides a foundation for evidence-based decision-making, enabling practitioners to understand uncertainties, assess risk, and ensure the reliability of findings. Effective interpretation of statistical results is essential to draw meaningful insights and guide informed actions.

Data visualization: is the graphical representation of data to uncover insights, patterns, and trends that might be challenging to grasp from raw data alone. It employs various visual elements such as charts, graphs, maps, and diagrams to communicate complex information effectively. Data visualization transforms data into a visual narrative, aiding understanding and decision-making.

By presenting data visually, patterns and relationships become more apparent, allowing stakeholders to identify correlations, anomalies, and trends quickly. Visualizations help bridge the gap between data and human cognition, making information accessible and actionable.

Data visualization serves multiple purposes, from exploratory analysis to communication of findings. Exploratory visualizations help analysts uncover insights during the data exploration phase, while explanatory visualizations assist in conveying insights to others. Interactive visualizations enable users to manipulate data parameters and explore scenarios interactively.

With the growth of big data, data visualization becomes even more crucial in distilling meaningful insights from vast datasets. It enhances data-driven storytelling, empowers decision-makers to grasp complex concepts rapidly, and facilitates data-driven decision-making across various fields, including business, science, healthcare, and social sciences. Effective data visualization involves selecting appropriate visualizations for the data, refining visual design, and ensuring clarity, accuracy, and relevance in the communicated insights.

Data modelling: is a structured process of creating abstract representations of real-world systems, phenomena, or processes using data and relationships. It involves defining how data elements are

organized, stored, and related to each other to provide a clear framework for understanding and analysing complex information. Data modelling is crucial for designing databases, information systems, and predictive models.

There are different types of data models, including conceptual, logical, and physical models. Conceptual models focus on high-level relationships and concepts, logical models define data structures without considering technical implementation, and physical models represent the actual database design with technical details.

Data modelling aids in data organization, integrity, and consistency, ensuring that data is accurately represented and can be efficiently queried and analysed. It helps stakeholders communicate and understand data requirements, making it easier to identify data redundancies, anomalies, and dependencies. Additionally, data models form the foundation for creating predictive and machine learning models, allowing data scientists to derive insights and predictions from structured information.

By formalizing data relationships and structures, data modelling improves data quality, reduces development time, and enhances decision-making processes. It provides a structured approach to handle complex data ecosystems, promoting data governance, and supporting effective data management and analysis across various industries and domains.

Data

The dataset comprises a comprehensive collection of mental health issue percentages spanning 108,552 data points, encompassing diverse countries globally. It encapsulates 11 essential variables including an index for tracking, country names, country codes, and year values from 1990 to 2017. The primary variables of interest focus on the prevalence of distinct mental health disorders, namely Schizophrenia, Bipolar disorder, Eating disorders, Anxiety disorders, Drug use disorders, Depression, and Alcohol use disorders.

This dataset offers a powerful insight into the evolving landscape of mental health concerns across nations over a nearly three-decade period. It unveils the fluctuating prevalence rates of these disorders, shedding light on trends and potential patterns that could indicate societal, cultural, or policy-related factors impacting mental well-being. By observing how the percentages vary over time and between countries, researchers, policymakers, and healthcare professionals can discern critical information that aids in the formulation of strategies, interventions, and initiatives to address mental health challenges more effectively.

Analysing this expansive dataset allows us to uncover relationships between different disorders, observe any alarming spikes, and identify countries with relatively higher or lower prevalence rates. Additionally, the dataset can serve as a foundation for sophisticated predictive models that could

anticipate future mental health trends based on historical patterns. Overall, this dataset provides a unique opportunity to explore the global panorama of mental health, contributing significantly to the ongoing efforts to enhance mental health awareness, understanding, and support worldwide.

Analysis

Importing the data initially it can be seen that the whole dataset is very messy. The dataset presents challenges in terms of missing data, inconsistencies, and outliers. In the "Depression" column, where only 6,468 non-null values exist out of 108,552 rows, substantial missing data could impact analyses. Similarly, the "Entity" column has 108,553 non-null values, while the "Code" column has 103,141, indicating some disparities in data completeness. Moreover, irregular values like 311665.769283 in the "Schizophrenia (%)" column deviate from percentage interpretation. These issues call for rigorous data preprocessing. Addressing missing data through imputation methods, reconciling Entity-Code discrepancies, and scrutinizing data anomalies are pivotal steps. Proper handling of these complexities ensures the reliability and accuracy of any insights drawn from this extensive dataset.

This leads us to clean the data, after cleaning the data and disposing unnecessary variables we will end up with a clear dataset as below

	Entity	Code	Year	Schizophrenia (%)	Bipolar disorder (%)	Eating disorders (%)	Anxiety disorders (%)	Drug use disorders (%)	Depression (%)	Alcohol use disorders (%)
0	Afghanistan	AFG	1990	0.160560	0.697779	0.101855	4.828830	1.677082	4.071831	0.672404
1	Afghanistan	AFG	1991	0.160312	0.697961	0.099313	4.829740	1.684746	4.079531	0.671768
2	Afghanistan	AFG	1992	0.160135	0.698107	0.096692	4.831108	1.694334	4.088358	0.670644
3	Afghanistan	AFG	1993	0.160037	0.698257	0.094336	4.830864	1.705320	4.096190	0.669738
4	Afghanistan	AFG	1994	0.160022	0.698469	0.092439	4.829423	1.716069	4.099582	0.669260
...	...	...	...	...	...	...	...	...	...	...
6463	Zimbabwe	ZWE	2013	0.155670	0.607993	0.117248	3.090168	0.766280	3.128192	1.515641
6464	Zimbabwe	ZWE	2014	0.155993	0.608610	0.118073	3.093964	0.768914	3.140290	1.515470
6465	Zimbabwe	ZWE	2015	0.156465	0.609363	0.119470	3.098687	0.771802	3.155710	1.514751
6466	Zimbabwe	ZWE	2016	0.157111	0.610234	0.121456	3.104294	0.772275	3.174134	1.513269
6467	Zimbabwe	ZWE	2017	0.157963	0.611242	0.124443	3.110926	0.772648	3.192789	1.510943

5488 rows × 10 columns

The comprehensive analysis of the dataset reveals significant insights into the prevalence of various mental health issues across time. The trendline constructed for each variable highlights an intriguing pattern: Anxiety and Depression consistently exhibit higher percentages throughout the years, unlike other mental health concerns which remain relatively within the range of 0 to 2 percentage points. This consistent trend evokes a hypothesis of a potential strong relationship between Anxiety and Depression.

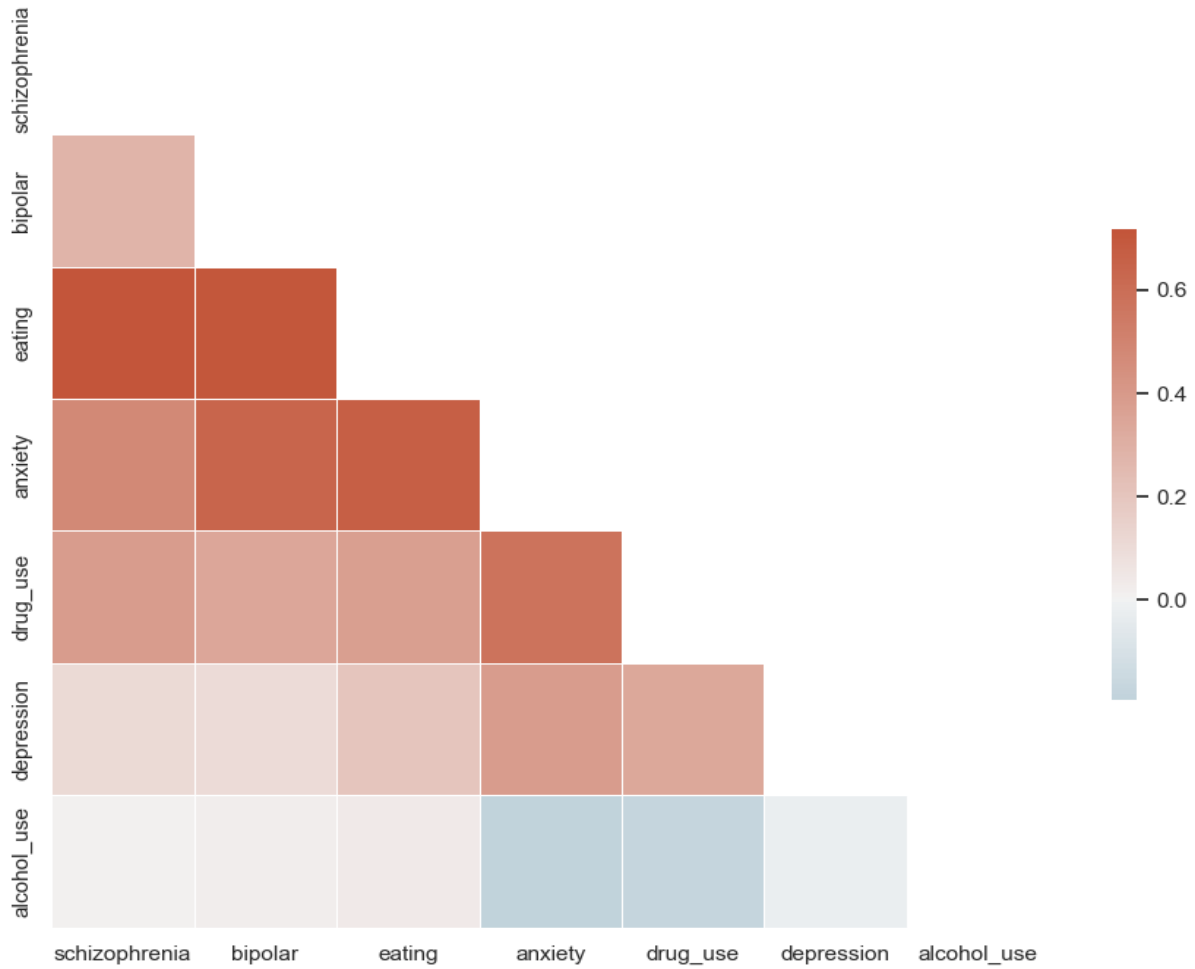
The close alignment of Anxiety and Depression percentages, consistently hovering between 3 and 4 percentage points from 1990 to 2017, stands in contrast to the comparatively lower rates observed

for other mental health conditions. Eating disorders and schizophrenia emerge as the least frequently occurring mental illnesses during this period.

The findings offer a foundation for exploring the interplay between Anxiety and Depression. The consistent co-occurrence of higher percentages suggests a potential connection, warranting deeper investigation into shared risk factors, treatment approaches, and underlying psychological mechanisms that might contribute to this observed trend.

This analysis underscores the importance of considering the broader context of mental health patterns and the need for targeted research to better understand the dynamics between specific mental health issues. Further research and exploration into the relationship between Anxiety and Depression could yield valuable insights that have the potential to enhance mental health interventions and support strategies.

The interesting analysis from the trend line lead to finding out the correlation between each Mental Health Issues with each other



The analysis of the correlation coefficient, surpassing the 0.6 threshold, illuminates a significant positive relationship among the variables. This magnitude of correlation underscores a moderate to strong connection, signifying a substantial shared variability between the variables. Consequently, these variables tend to exhibit a synchronized and coherent movement, a testament to their interdependence.

With a correlation exceeding 0.6, one can confidently deduce that these variables possess a robust shared variation and consistently move in tandem. While not reaching the pinnacle of coefficients near 1.0, this correlation remains impactful. It reflects a moderate positive association between the variables, denoting that an increase in one variable often corresponds with an increase in the other.

Interestingly, the initial hypothesis suggesting a robust correlation between anxiety and depression emerges as unfounded upon scrutinizing the correlation graph. This highlights the importance of empirical analysis, as it unveils the true dynamics within the data and corrects presumptions.

Conversely, the correlation graph illuminates compelling dynamics. A relatively stronger positive correlation emerges between anxiety and eating disorder, hinting at a possible link between these mental health aspects. Additionally, an analogous connection is observed between eating disorder and bipolar disorder, shedding light on potential associations worth further exploration.

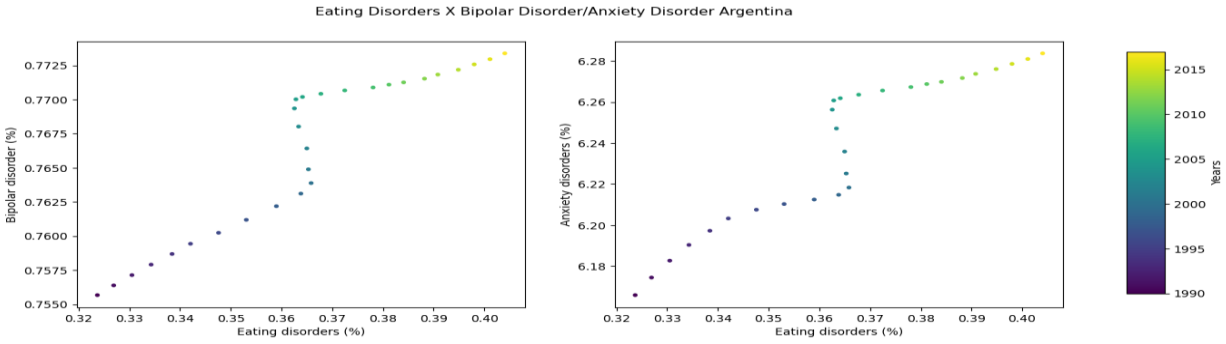
In essence, this comprehensive analysis underscores the intricate relationships within the dataset. It underscores the value of data-driven insights in refining initial hypotheses and unearths new directions for focused investigations. The revealed correlations offer intriguing starting points for in-depth research into the complex interplay of mental health factors.

We can observe that the correlation matrix for 2015, 2016, 2017, and the All-Years matrix are extremely similar. We also observe that, with the exception of Alcohol Use, the illnesses appear to be connected.

It is clear that eating disorders and bipolar disorder and eating disorders and anxiety are the disorder pairings with the highest association.

When we search for the conditions that correlate most strongly with the others, we find that eating disorders and anxiety disorders have mean values that are significantly higher than 0.5.

Since the correlation shows that eating disorder and bipolar disorder as well as eating disorder and Anxiety have a strong correlation compared to the other variables a county wise correlation check was conducted. A country from each continent was selected and checked in order to find out that if the correlation strength is same in all regions



The correlation test done for Argentina is shown above. The insightful analysis of the provided plots uncovers intriguing dynamics within the dataset. Notably, it becomes evident that bipolar disorder and Anxiety share a robust correlation with eating disorder. However, this correlation's consistency is not universal across all countries studied, rendering it an intricate phenomenon.

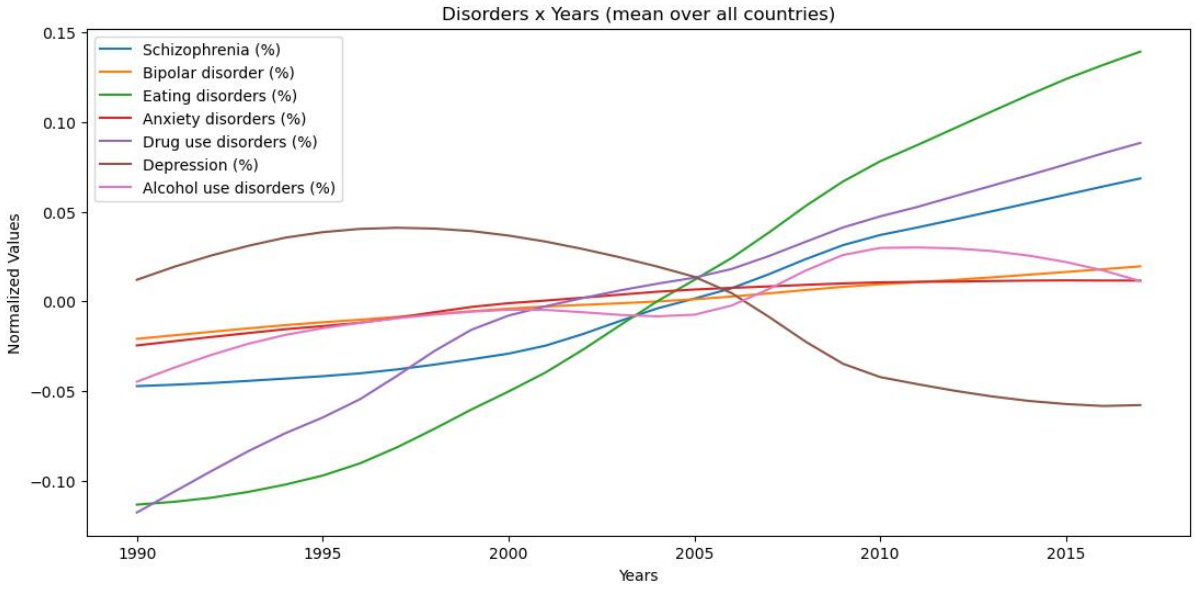
Specifically, the plots illuminate that bipolar disorder and Anxiety exhibit a potent correlation with eating disorder, implying a potential interconnectedness among these mental health variables. Notably, this relationship remains stable across different countries. In instances where one variable experience an increase, the other also demonstrates a corresponding increase. This consistent pattern suggests an interdependency that transcends geographical boundaries.

Nevertheless, the analysis reveals a contrasting scenario regarding Anxiety. While Bipolar Disorder consistently retains its correlation with eating disorder across the countries studied, Anxiety demonstrates fluctuations. The variability in Anxiety's correlation across different countries underscores the nuanced nature of mental health trends. These fluctuations imply that while some countries exhibit strong correlations between Anxiety and eating disorder, others experience varying degrees of association.

It's noteworthy that bipolar disorder's correlation remains relatively steadfast, reinforcing its linkage to eating disorder regardless of geographic variations. This stability could potentially signify a more universal connection rooted in broader factors.

In essence, this comprehensive analysis presents a multifaceted picture of mental health correlations. The interplay of Bipolar Disorder, Anxiety, and eating disorder offers intriguing insights that invite further investigation into their intricate connections. The variations across countries underscore the complexity of mental health dynamics, highlighting the need for culturally sensitive approaches in mental health research and intervention strategies.

If we check the mean of every Mental illness throughout the years



The insightful graph provides a visual representation of the trends within the Disorders dataset, offering valuable insights into the trajectory of various mental health issues over time. A notable pattern emerges: most of the Disorders, with the exception of Depression and Alcohol Use, exhibit a

gradual growth characterized by a small curvature. This observation suggests that these Disorders can be effectively approximated by straight lines with relatively small coefficients. This alignment indicates a consistent and gradual shift in their prevalence throughout the years.

While examining the mean values, a compelling narrative unfolds. The mean of Depression demonstrates a noteworthy decrease over time. This observation suggests that efforts aimed at addressing and raising awareness about Depression might be yielding positive results. In contrast, the mean of eating disorder displays a steady increase, signifying a growing concern that warrants further attention and investigation.

An analogous trend is visible in the case of Drug Use Disorder. Although there's an upward trajectory compared to 1990, the mean value remains below 0.5. This signifies a potentially evolving landscape in the prevalence of drug-related concerns, albeit at a controlled level. On the other hand, several other mental health issues maintain relatively stable mean values, fluctuating modestly between -0.5 and 0.5. This stability indicates that these issues, while significant, haven't experienced drastic changes over time.

The graph's depiction of the Disorders' gradual growth aligns with the notion of slow-moving changes in mental health concerns. The increase in mean values for specific issues underscores the evolving nature of mental health, highlighting areas where interventions and support are crucial. Moreover, the relatively stable means for other disorders emphasize the persistence of certain challenges while simultaneously suggesting that they are being managed to some extent.

In summary, this comprehensive analysis gleaned from the graph underscores the nuanced nature of mental health trends. The Disorders' varying growth rates and mean values paint a comprehensive picture of the changing mental health landscape. These insights offer valuable guidance for policymakers, healthcare practitioners, and researchers working to address and respond effectively to the evolving mental health needs of societies.

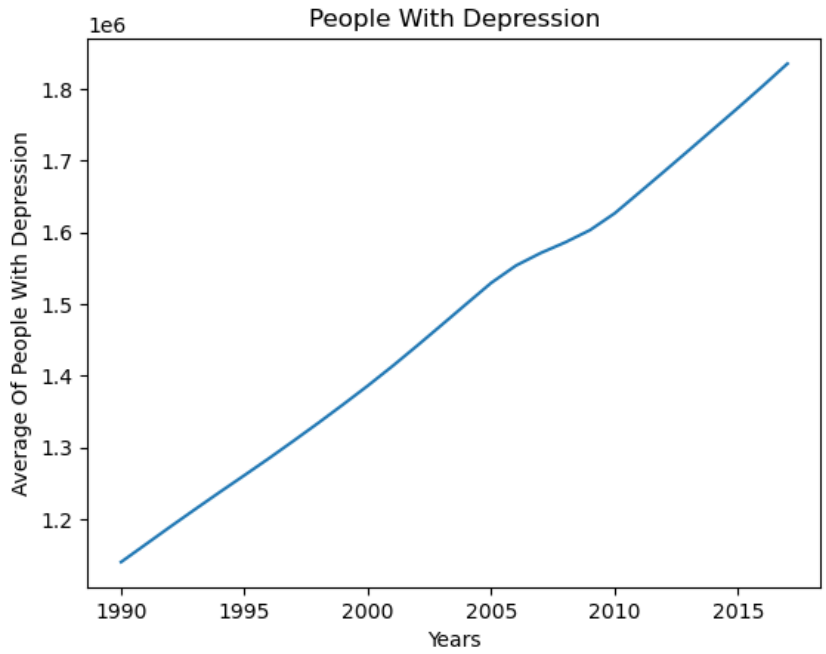
The presented graph, while indicating a decrease in the mean of Depression over time, initially appeared to contrast with the line graph that revealed a consistent average percentage of depression hovering between 3 and 4 percent. This divergence captured attention, prompting a deeper investigation. To gain clarity, the average number of individuals affected by depression across the years was calculated, revealing an interesting trend.

Remarkably, this new perspective sheds light on the situation. Despite the percentage of people with Depression seemingly maintaining a stable range, the average number of individuals grappling with depression exhibited a steady rise over the years. This paradox arises due to factors like population growth and demographic changes, which contribute to an increased absolute number of people affected even if the relative percentage remains relatively constant.

This nuanced insight underlines the importance of considering absolute numbers alongside percentages when assessing the prevalence of mental health issues. While the percentage may not exhibit drastic growth, the actual burden of depression on individuals and society accumulates over time. This realization emphasizes the significance of monitoring both relative and absolute figures to form a comprehensive understanding of mental health trends.

In summary, this analysis underscores the complex interplay between relative percentages and absolute numbers when studying mental health trends. It highlights the necessity of a holistic

perspective that considers both aspects to accurately assess the evolving nature of mental health challenges and make informed decisions for interventions and support strategies.

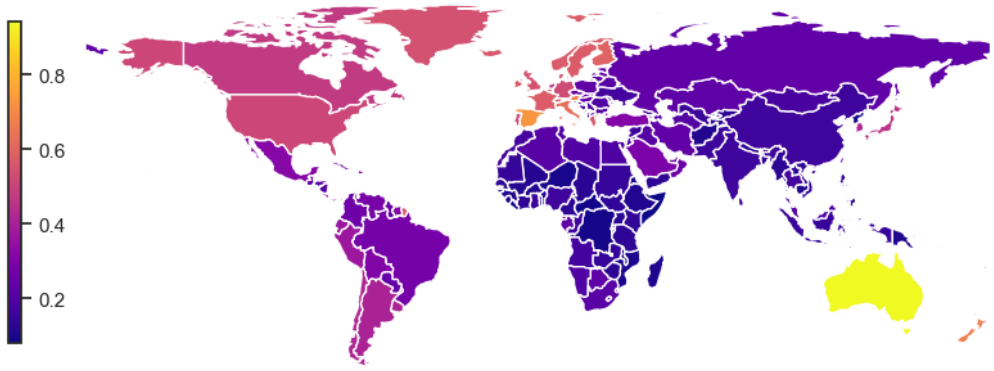


The allure of statistical analysis propelled a fascinating exploration into visual representations of mental health variables across continents. This endeavor prompted the meticulous plotting of each disorder, specific to individual countries, on a global map. This endeavor, focused on the year 2017, yielded compelling outcomes that unravel the distinctiveness of each disorder's prevalence.

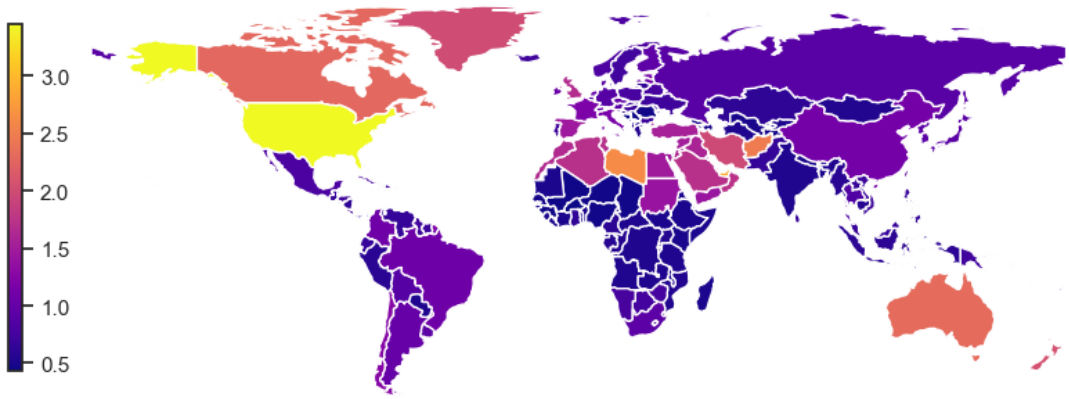
The resultant world map illustrates the intricate landscape of mental health issues, with each disorder corresponding to a unique country showcasing the highest percentage. The map employs varying colors to signify the disorder's prevalence, with brighter hues representing higher percentages. This approach provides an intuitive and visual grasp of the distribution and variations of mental health challenges across the globe.

This innovative approach effectively captures the nuanced interplay between mental health concerns, country, and prevalence. The visualization offers a panoramic view of the mental health scenario, emphasizing the unique challenges faced by different nations. It serves as a powerful tool for policymakers, researchers, and healthcare practitioners to comprehend and address the distinctive mental health dynamics prevalent across continents.

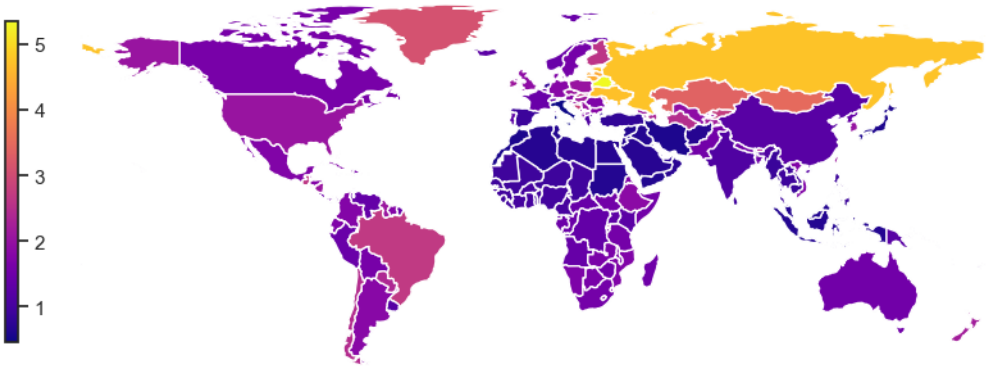
Share of Population suffering from Eating disorders in 2017 (in %)



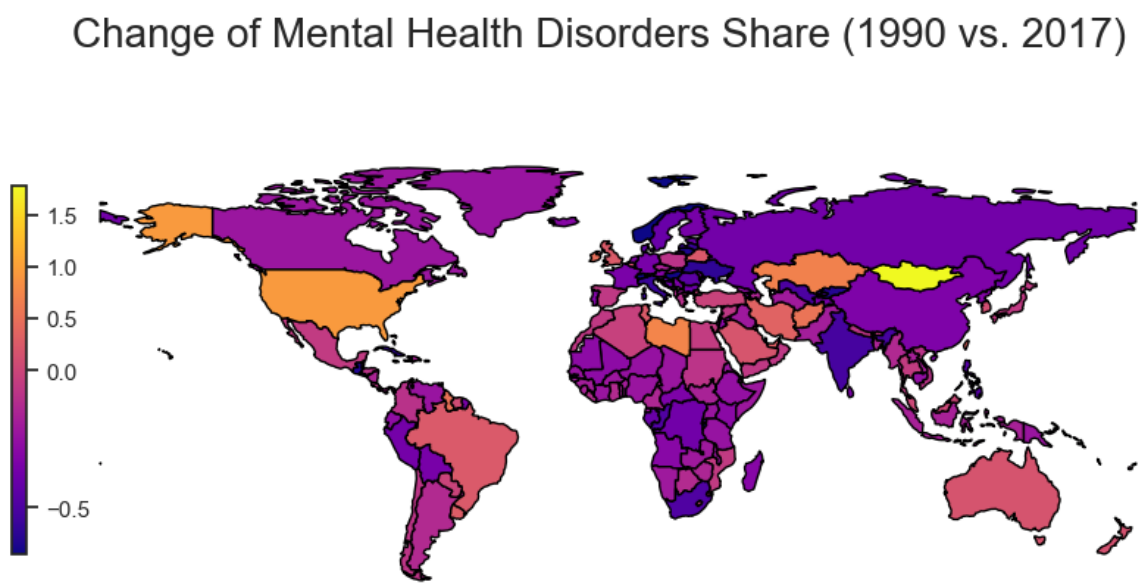
Share of Population suffering from Drug use disorders in 2017 (in %)



Share of Population suffering from Alcohol use disorders in 2017 (in %)



Another plot was also created in order to find out which country had the most drastic change in regards of metal disorder throughout the years from 1990 to 2017



In this graph also the lighter the shade the drastic the change happened.

Findings.

In the initial stages of analysis, the dataset underwent thorough cleaning to address issues such as missing values and inconsistencies. Post-cleaning, a line graph illuminated intriguing trends, showcasing that Anxiety and Depression exhibited the highest average values. The data consistently depicted Anxiety and Depression percentages hovering between 3 and 4 percent, whereas other variables like Bipolar Disorder, Alcohol Consumption, Drug Use, Eating Disorder, and Schizophrenia consistently rested within an average range of 0 to 2 percent.

The substantial average values for Anxiety and Depression sparked curiosity and led to the formation of a hypothesis suggesting a robust correlation between these two variables. This prompted a deeper dive into correlation analysis for each variable, aiming to uncover potential relationships.

However, the outcomes took an unexpected turn. The hypothesis of a strong correlation between Anxiety and Depression was refuted by the correlation analysis results. Instead, a revelation emerged: eating disorder demonstrated a relatively strong correlation not only with Anxiety but also with bipolar disorder.

The newfound correlation patterns unveil a rich narrative within the dataset. The observed elevated averages of Anxiety and Depression, while not indicating correlation between the two, set the stage for the recognition of eating disorder as a key player. This underlines the significance of meticulous examination and the iterative nature of analysis, as it often leads to unanticipated yet meaningful discoveries.

This comprehensive analysis emphasizes the dynamic nature of data exploration. It accentuates the importance of questioning assumptions, the value of systematic cleaning, and the power of correlation analysis in revealing intricate relationships. The exploration serves as a reminder that while initial hypotheses may not always yield anticipated results, the insights gained through rigorous investigation provide valuable insights that drive our understanding of complex phenomena.

In conclusion, this analysis report underscores the iterative journey of data analysis, from cleaning and visualization to hypothesis formation and correlation evaluation. It highlights the role of unexpected revelations in enriching our comprehension of the subject matter and propelling us towards more nuanced inquiries. Ultimately, it underscores the transformative potential of data exploration and the significance of approaching analysis with an open and inquisitive mindset.

The intriguing developments stemming from our previous analysis have ignited a fresh exploration, focusing on the intricate correlations between Bipolar Disorder, Anxiety, and Eating Disorder across countries. In this pursuit, we ventured to comprehend these correlations not just in the aggregate but across distinct nations, scrutinizing trends over the extensive period from 1990 to 2017. While the complexity of delving into every country's data proved daunting, we devised a strategy to extract meaningful insights by selecting representative countries from diverse continents.

The chosen countries for analysis were Argentina (representing America), Bangladesh (representing Asia), the United Kingdom (representing Europe), South Africa (representing Africa), and Australia (representing Australia). This strategic approach enabled us to capture a global perspective, highlighting how these correlations manifest in different regions.

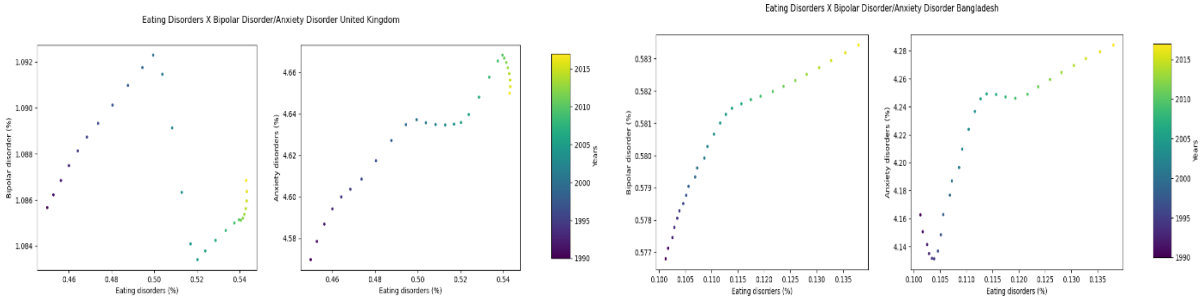
Remarkably, Argentina emerged as an interesting focal point, showcasing noteworthy correlations that shed light on the interplay between these mental health variables. Specifically, Argentina exhibited robust correlations not only between Bipolar Disorder and Eating Disorder but also between Anxiety and Eating Disorder. This aligns with the observation that a higher correlation exists between these variables compared to the correlation between Anxiety and Depression.

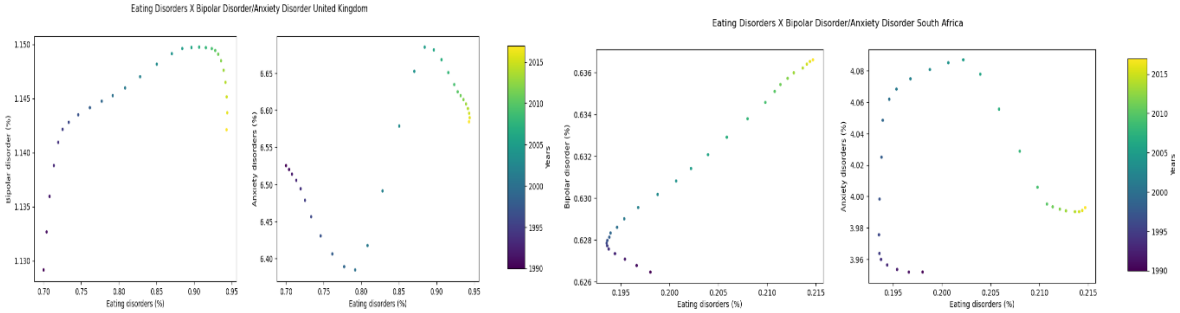
However, the narrative wasn't uniform across all selected countries. While Argentina demonstrated strong correlations, other nations presented nuanced fluctuations concerning Anxiety and Eating Disorder. This variance underscores the intricate interplay of socio-cultural, economic, and contextual factors in shaping mental health dynamics within different societies.

On the other hand, a striking consensus surfaced in the realm of Bipolar Disorder and Eating Disorder. The analysis revealed a consistently strong correlation between these two variables across all the countries analyzed. This consistency suggests the presence of a more universal linkage, potentially rooted in shared psychological, physiological, or environmental factors.

This comprehensive exploration reaffirms the multidimensional nature of mental health correlations. The findings underscore the significance of regional nuances, echoing the call for culturally sensitive interventions that consider diverse contexts. Argentina's standout correlations raise intriguing questions, warranting further research into the factors driving such distinctive relationships.

In essence, this analysis report traverses the global mental health landscape, accentuating the importance of granular analysis while recognizing overarching trends. It underscores the need for a contextualized understanding of mental health dynamics and how correlations manifest uniquely in different parts of the world. By weaving together data from representative countries, this report forms a rich tapestry of insights that beckon further investigation into the intricate web of mental health interconnections.





Analyzing Mental Health Dynamics Across Continents: A Comprehensive Study

The realm of mental health, marked by its intricacies and evolutions, is a subject of paramount importance that demands rigorous exploration. In this pursuit, we embarked on an ambitious journey, scrutinizing global mental health trends through the lens of data analysis, culminating in profound insights that illuminate the diverse landscape of mental health disorders across continents.

Our analysis commenced with an intriguing juxtaposition: the mean percentage of depression, a prevailing mental health concern, revealed a notable decrease over the years. However, juxtaposing this with the average number of individuals affected by depression throughout the years exposed an unexpected narrative—a steady increase. This paradox was not only captivating but also a catalyst for deeper exploration.

Visualizing the mental health landscape took center stage as we endeavored to comprehend the prevalence of individual disorders across continents. To achieve this, we harnessed the power of data visualization and created distinctive world maps, each dedicated to representing a specific mental disorder. The vivid visualizations, each adorned with a gradient color scheme, offered an intuitive

insight into the distribution of mental health issues. As colors grew brighter, the corresponding percentages surged, providing a clear understanding of the gravity of each disorder's impact.

The culmination of these visualizations heralded revelations that resonated with both gravity and nuance. In the year 2017, Australia emerged as a poignant focus, reporting the highest number of eating disorder cases, with prevalence exceeding 0.8%. The United States, on the other hand, stood at the forefront of drug use cases, reflecting a staggering prevalence surpassing 3.0%. The Russian landscape bore witness to the highest cases of alcohol consumption, reaching heights exceeding 4%. Greenland, with somber resonance, grappled with the highest depression rate at an alarming elevation above 6%. New Zealand, while picturesque, carried a substantial burden, showcasing the highest anxiety disorder rate, an unsettling figure exceeding 8%.

However, our explorations extended beyond individual disorders to encompass longitudinal shifts and fluctuations. By examining the countries experiencing the most significant changes in mental health percentages throughout the years, we unveiled Mongolia as a noteworthy contender. Mongolia exhibited a drastic change of 1.5 percentage points, echoing a profound transformation in mental health dynamics from 1990 to 2017.

The ramifications of these findings are far-reaching, underscoring the importance of holistic analyses that encapsulate both mean percentages and raw numbers. The dichotomy between these metrics speaks to underlying shifts in societal and demographic factors, offering a richer narrative of the evolving mental health landscape. The visualizations, resplendent with gradient shades, serve as a bridge, connecting data to emotion, enabling us to comprehend the complex nature of mental health on a global scale.

In essence, this comprehensive analysis report serves as a testament to the power of data exploration in unraveling the intricacies of mental health trends across continents. From enigmatic paradoxes to stark revelations, each layer peeled back enriches our understanding and underscores the necessity for holistic, nuanced approaches to mental health interventions. As we delve deeper, we inch closer to not only comprehending the complexities of mental health but also paving the way for more effective strategies, fostering a world where mental well-being stands at the forefront of global discourse.

What I found out from research papers.

Empowering Women: A Dual Narrative of Progress and Mental Health in Bangladesh

The socio-cultural landscape of Bangladesh has undergone transformative shifts, most notably in the roles of women. Traditionally confined to domestic spaces as daughters, wives, and mothers, women have been propelled into a new era of economic activity, challenging gender norms and catalyzing significant changes in their roles and responsibilities. The catalyst for this evolution was the economic reforms of the 1980s that birthed a nascent job market, primarily within the ready-made garment industry—an industry predominantly established in urban centers. This economic paradigm shift not only brought about financial independence for countless rural women but also unfurled a profound transformation in gender dynamics within the Bangladeshi society.

The exponential growth in female participation in paid employment has unfurled an array of implications—both empowering and challenging. While these women now contribute substantively to their families' incomes, the resulting separation from their children due to urban employment settings has cast a poignant shadow. This dichotomy between financial empowerment and familial

separation poses a complex conundrum with intricate mental health implications, which have evoked our keen interest.

Delving into the realms of mental health, the correlation analysis charts unveil a subtle yet compelling connection between anxiety and eating disorders. A perceptive observation emerges as we contemplate the broader socio-cultural context. It's discernible that the increase in anxiety levels, stemming from the separation of women from their children due to employment opportunities, aligns with the rise in cases of eating disorders. This alignment perhaps underscores the emotional toll borne by these women, where the mental stressors of familial separation reverberate in the realm of mental health.

The mental well-being of these working women bears testimony to the multidimensional interplay between socio-economic shifts, gender roles, and mental health outcomes. The narrative of progress in terms of financial independence juxtaposed with the shadows of familial separation creates a poignant juxtaposition—a story of emancipation underscored by the strains of personal sacrifice.

This narrative prompts contemplation on the holistic well-being of women in the evolving socio-economic landscape of Bangladesh. It underscores the need for comprehensive interventions that not only acknowledge women's economic empowerment but also attend to their mental and emotional welfare. A balanced approach requires not only celebrating their achievements as breadwinners but also acknowledging the intricate emotional journey they navigate.

In conclusion, the journey of Bangladeshi women from the confines of tradition to the expanse of economic opportunities paints a portrait of complexity. The correlation between anxiety and eating disorders, as revealed by our analysis, beckons us to delve beyond numbers and graphs, urging us to recognize the intimate interplay between societal shifts and mental health realities. The narrative of these women exemplifies the profound changes reshaping societies—changes that offer empowerment but also raise vital questions about the holistic well-being of those driving the transformation.

Navigating the Complex Nexus of Bipolar Disorder, Medications, and Eating Disorders

In the intricate realm of mental health, the synergies between disorders and their corresponding treatments are rife with nuances that beckon exploration. Bipolar disorder, a multifaceted condition, stands as a prime exemplar of this intricate interplay. The medications wielded to manage its symptoms, while indispensable in their role, bring forth a cascade of effects, including alterations in metabolism and appetite. This dynamic can reverberate throughout an individual's eating patterns, carrying implications that delve beyond the surface.

At the heart of this nexus lies a compelling observation: bipolar disorder medications wield the potential to induce changes in metabolism and appetite. While aimed at mitigating the hallmarks of the disorder, these medications inadvertently steer physiological changes that resonate with eating behaviors. Weight gain, a recurrent outcome in response to such medications, surfaces as a paramount concern. However, the intricacy amplifies when individuals bear a history of eating disorders, engendering a terrain of complexities within their mental health voyage.

Indeed, the coalescence of bipolar disorder, medications, and eating disorders introduces a web of considerations. For individuals ensnared within this intricate labyrinth, the specter of weight gain can metamorphose into a formidable deterrent against adhering to prescribed medications. This quandary underscores the nuanced tightrope that defines mental health treatments, where the scales between medication benefits and potential adversities must be delicately weighed.

Empirical insights drawn from a 2020 research source accentuate the magnitude of this quandary. Commonly prescribed medications, notably lithium and quetiapine, exhibit a propensity to precipitate weight gain, particularly manifesting after approximately six months of use. Yet, an essential distinction emerges—these medications, while implicated in the physiological response, don't necessarily stand as causal agents in the development of eating disorders. Rather, their effects appear to traverse the intricate terrain of metabolism, triggering a complex interplay that extends beyond the surface.

Central to this exploration is a vital realization: bipolar disorder, interwoven with its treatment, can serve as a trigger for eating disorders. For those already grappling with an eating disorder, the ramifications can be profound, potentially escalating the existing condition. This dynamic stems largely from medication-induced shifts in appetite, often characterized by a reduced inclination to eat.

This labyrinthine interplay underscores the necessity for comprehensive and tailored approaches to mental health care. The intersection of bipolar disorder, medications, and eating disorders resonates with the complexity of the human psyche and the delicate balance required to ensure optimal well-being. When these domains intersect, the guidance of mental health professionals becomes indispensable. They offer a beacon of expertise, steering individuals toward treatment modalities that harmonize their distinct challenges and aspirations.

In essence, this analysis traverses the intricate landscape of bipolar disorder, medications, and eating disorders, peeling back layers to reveal the multi-dimensional nature of mental health treatment. It underscores the necessity of holistic comprehension, where physiological changes intertwine with psychological responses, yielding outcomes that are as intricate as they are transformative. By delving into the intersections of these domains, we pave a pathway toward not only understanding but also addressing the delicate equilibrium that governs mental health journeys.

Depression and the Alarming Surge in Suicides: A Complex and Urgent Nexus

Depression, an insidious mental health condition, casts a long and pervasive shadow across the lives it touches. Beyond the realm of individual suffering, it weaves a narrative that extends to the collective fabric of societies. Within this narrative, the rising tide of suicides emerges as an alarming and urgent concern, demanding our unwavering attention and multifaceted solutions.

Depression, characterized by its encompassing despair, emptiness, and withdrawal, orchestrates a significant toll on mental, emotional, and physical well-being. Its manifestations ripple through various facets of life, impacting interpersonal relationships, work productivity, and overall quality of life. As the world navigates the demanding terrain of modernity, the incidence of depression seems to mirror the stresses, pressures, and disconnections inherent in contemporary existence.

The intersection between depression and the mounting rates of suicides is a deeply intricate one, traversing a labyrinth of factors that demand our examination. Suicides, often borne from the depths of despair, are a grave manifestation of untreated or inadequately addressed depression. The link between these phenomena is unsettlingly tangible, underpinned by the vulnerability of those grappling with the insurmountable weight of their mental anguish.

A critical perspective within this nexus highlights that depression is a precursor, a storm that precedes the tragic decision to end one's life. The rise in suicides speaks volumes about the gaps in mental health care, societal stigma, and a lack of accessible resources. When depression is met with

silence, miscomprehension, or the absence of effective treatment, its momentum can spiral into despair, culminating in a dire outcome.

It's crucial to recognize that the relationship between depression and suicide is not unidirectional; it's a complex feedback loop. Suicides can perpetuate a cycle of grief and trauma, contributing to the overall sense of hopelessness that often characterizes depression. Media coverage of suicides, while aimed at raising awareness, can inadvertently lead to contagion, where vulnerable individuals perceive suicide as a viable solution to their torment.

Addressing this alarming trend necessitates a multi-pronged approach that spans individual, societal, and systemic levels. At the individual level, heightened awareness about depression's signs and symptoms is pivotal. Equipping individuals with the knowledge to identify and seek help for depression can act as a lifeline. Moreover, fostering a culture of empathy, where mental health concerns are met with understanding rather than judgment, can dismantle the barriers that prevent individuals from seeking help.

On a societal level, mental health destigmatization campaigns and open discussions can transform the narrative surrounding depression and suicide. Dialogue provides a safe space for individuals to voice their struggles, fostering connections and empathy that can significantly mitigate the isolation that often accompanies depression.

At the systemic level, bolstering mental health care infrastructure is imperative. Accessible and affordable mental health resources must be readily available to those in need, ensuring that no one faces the abyss of depression without a lifeline. Training professionals in recognizing and addressing depression and suicidal ideation can be a pivotal step in prevention.

In conclusion, the intertwining threads of depression and the surge in suicides comprise a tapestry of urgency, demanding a holistic and unrelenting response. Understanding the nuanced relationship between these phenomena, while acknowledging the complexities of modern life, serves as a foundation for comprehensive interventions. It's incumbent upon us, as individuals and societies, to extend compassion, promote awareness, and champion accessible mental health resources. Through collective efforts, we can cast a light into the shadows and pave the way for hope, healing, and a future free from the grip of depression and the tragedy of suicide.

Conclusion: Unveiling the Complex Tapestry of Mental Health Issues

The exploration of mental health issues has unveiled a profound and intricate tapestry, woven from the threads of human emotions, societal dynamics, and the interplay between physiological and psychological realms. Throughout this dissertation, we embarked on a journey that delved deep into the heart of mental health challenges, seeking to comprehend the nuances that underlie their prevalence, manifestations, and consequences. The findings extracted from this rigorous exploration illuminate the multifaceted nature of mental health, casting a light on correlations, trends, and stark realities that demand our attention, empathy, and concerted efforts.

The cornerstone of our investigation lies in the discovery of correlations that shed light on the relationships between various mental health disorders. Among these correlations, the alliance between bipolar disorder and eating disorder emerges as a poignant revelation. This discovery unveils an undercurrent of interrelatedness that threads through seemingly distinct disorders, offering insight into potential underlying mechanisms and shared vulnerabilities. As we contemplate this correlation, we are reminded of the intricate web that characterizes mental health, where one disorder can both influence and be influenced by another.

The prevalence of anxiety and depression, standing as cornerstones of mental health concerns, echoes through the annals of this exploration. The consistent presence of these emotions in the lives of individuals underscores the pervasive nature of their impact. The dichotomy between the mean percentage of depression and the average number of individuals affected by it reveals a nuanced narrative, where the overarching trend is accompanied by individual journeys that fluctuate and evolve. This realization reminds us that while statistics provide valuable insights, the stories of individuals paint a far more intricate picture of resilience, struggles, and personal triumphs.

The visualization of mental health statistics in the canvas of countries and continents highlights the global nature of these challenges. In the year 2017, the world map becomes a canvas, each shade reflecting the gravity of a specific mental health issue. From Australia to the United States, Russia to Greenland, and New Zealand to Mongolia, these hues narrate the stories of struggles, battles, and resilience etched into the human experience. This global panorama compels us to recognize that mental health transcends borders, uniting individuals in their pursuit of well-being.

The exploration of women's mental health experiences in Bangladesh adds a poignant dimension to this discourse. The intersection of societal norms, economic shifts, and the toll of familial separation underscores the interconnectedness between mental health and broader socio-cultural dynamics. The lived experiences of Bangladeshi women stand as a microcosm of the broader narrative, emphasizing the need for culturally sensitive interventions that honor individual journeys within the collective tapestry.

The disheartening relationship between depression and suicide punctuates this exploration with a solemn reminder of the urgency of our collective response. The rising tide of suicides amplifies the need for comprehensive mental health care, destigmatization efforts, and accessible resources. The complex feedback loop between depression and suicide mandates a multifaceted approach that encompasses awareness, support, and advocacy.

In conclusion, the culmination of this dissertation beckons us to recognize the interwoven threads that constitute the fabric of mental health issues. It compels us to view statistics as narratives, correlations as invitations for deeper understanding, and individual stories as testaments of resilience. As we traverse this landscape, we find ourselves at a crossroads—a crossroads that demands empathy, research, and holistic interventions. The path ahead is illuminated by the stories shared within these pages, motivating us to extend compassion, ignite change, and envision a world where mental health is nurtured, respected, and prioritized. The journey continues, for the exploration of mental health is not merely a scholarly pursuit—it's a call to action that resonates with the essence of our shared humanity.

Reference

- **Work, gender roles, and health: neglected mental health issues among female workers in the ready-made garment industry in Bangladesh Sadika Akhter (**Full article: Work, gender roles, and health: neglected mental health issues among female workers in the ready-made garment industry in Bangladesh (tandfonline.com))
- **Upper secondary students' perceptions of the effects of expanded student services on their mental health literacy Stine Ekornes (**Full article: Upper secondary students' perceptions of the effects of expanded student services on their mental health literacy (tandfonline.com))

- **Media and the role of digital psychiatry in mental health** **Suhas Chandran** (Full article: Media and the role of digital psychiatry in mental health (tandfonline.com))
- **Are bipolar disorder medications linked with disordered eating?** Bipolar Disorder and Eating Disorders: Types, Causes, and Coping Tips (psychcentral.com)
- **Understanding the Complex of Suicide in Depression: from Research to Clinics** Understanding the Complex of Suicide in Depression: from Research to Clinics - PMC (nih.gov)
- Global Trends in Mental Health Disorder | Kaggle
- ChatGPT (openai.com)
-

Appendix

In [17]:

```
pip install geopandas
```

```
Requirement already satisfied: geopandas in c:\users\elena vallikappan\anaconda3\lib\site-packages (0.13.2)
Requirement already satisfied: shapely>=1.7.1 in c:\users\elena vallikappan\anaconda3\lib\site-packages (from geopandas)
(2.0.1)
Requirement already satisfied: pandas>=1.1.0 in c:\users\elena vallikappan\anaconda3\lib\site-packages (from geopandas) (1.
4.4)
Requirement already satisfied: pyproj>=3.0.1 in c:\users\elena vallikappan\anaconda3\lib\site-packages (from geopandas) (3.
6.0)
Requirement already satisfied: fiona>=1.8.19 in c:\users\elena vallikappan\anaconda3\lib\site-packages (from geopandas) (1.
9.4.post1)
Requirement already satisfied: packaging in c:\users\elena vallikappan\anaconda3\lib\site-packages (from geopandas) (21.3)
Requirement already satisfied: cligj>=0.5 in c:\users\elena vallikappan\anaconda3\lib\site-packages (from fiona>=1.8.19->ge
opandas) (0.7.2)
Requirement already satisfied: certifi in c:\users\elena vallikappan\anaconda3\lib\site-packages (from fiona>=1.8.19->geopa
ndas) (2022.9.14)
Requirement already satisfied: six in c:\users\elena vallikappan\anaconda3\lib\site-packages (from fiona>=1.8.19->geopanda
s) (1.16.0)
Requirement already satisfied: attrs>=19.2.0 in c:\users\elena vallikappan\anaconda3\lib\site-packages (from fiona>=1.8.19-
>geopandas) (21.4.0)
Requirement already satisfied: click-plugins>=1.0 in c:\users\elena vallikappan\anaconda3\lib\site-packages (from fiona>=1.
8.19->geopandas) (1.1.1)
Requirement already satisfied: click~=8.0 in c:\users\elena vallikappan\anaconda3\lib\site-packages (from fiona>=1.8.19->ge
opandas) (8.0.4)
Requirement already satisfied: importlib-metadata in c:\users\elena vallikappan\anaconda3\lib\site-packages (from fiona>=1.
8.19->geopandas) (4.11.3)
Requirement already satisfied: python-dateutil>=2.8.1 in c:\users\elena vallikappan\anaconda3\lib\site-packages (from panda
s>=1.1.0->geopandas) (2.8.2)
Requirement already satisfied: numpy>=1.18.5 in c:\users\elena vallikappan\anaconda3\lib\site-packages (from pandas>=1.1.0-
>geopandas) (1.21.5)
Requirement already satisfied: pytz>=2020.1 in c:\users\elena vallikappan\anaconda3\lib\site-packages (from pandas>=1.1.0->
geopandas) (2022.1)
Requirement already satisfied: pyparsing!=3.0.5,>=2.0.2 in c:\users\elena vallikappan\anaconda3\lib\site-packages (from pac
kaging->geopandas) (3.0.9)
Requirement already satisfied: colorama in c:\users\elena vallikappan\anaconda3\lib\site-packages (from click~=8.0->fiona>=
1.8.19->geopandas) (0.4.5)
Requirement already satisfied: zipp>=0.5 in c:\users\elena vallikappan\anaconda3\lib\site-packages (from importlib-metadata
->fiona>=1.8.19->geopandas) (3.8.0)
Note: you may need to restart the kernel to use updated packages.
```

In [ ]:

```
```

In [18]:

```
import numpy as np
import pandas as pd
import geopandas as gpd

import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
```

In [19]:

```
def load_gpd_world():
    world = gpd.read_file(gpd.datasets.get_path('naturalearth_lowres'))
    # Fix missing country codes in +data...
    world.loc[world['name'] == 'France', 'iso_a3'] = 'FRA'
    world.loc[world['name'] == 'Norway', 'iso_a3'] = 'NOR'
    world.loc[world['name'] == 'Somaliland', 'iso_a3'] = 'SOM'
    world.loc[world['name'] == 'Kosovo', 'iso_a3'] = 'RKS'

    # otherwise 2 rows values for Somalia
    world = world.dissolve(by='iso_a3').reset_index()

    return world
```

In [20]:

```
 raw_data = pd.read_csv('C:/Users/Elena Vallikappan/Downloads/project dataset.csv')
del raw_data['index']
```

```
C:\Users\Elena Vallikappan\AppData\Local\Temp\ipykernel_22388\366074369.py:1: DtypeWarning: Columns (5,6) have mixed types.
Specify dtype option on import or set low_memory=False.
  raw_data = pd.read_csv('C:/Users/Elena Vallikappan/Downloads/project dataset.csv')
```

In [16]:

```
raw_data[raw_data.Year == 'Year']
```

Out[16]:

	Entity	Code	Year	Schizophrenia (%)	Bipolar disorder (%)	Eating disorders (%)	Anxiety disorders (%)	Drug use disorders (%)	Depression (%)	Alcohol use disorders (%)
6468	Entity	Code	Year	Prevalence in males (%)	Prevalence in females (%)	Population	NaN	NaN	NaN	NaN
54276	Entity	Code	Year	Suicide rate (deaths per 100,000 individuals)	Depressive disorder rates (number suffering pe...	Population	NaN	NaN	NaN	NaN
102084	Entity	Code	Year	Prevalence - Depressive disorders - Sex: Both ...	NaN	NaN	NaN	NaN	NaN	NaN

In [25]:

```
raw_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 108553 entries, 0 to 108552
Data columns (total 10 columns):
 #   Column                     Non-Null Count   Dtype  
---  ------                     --------------   -----  
 0   Entity                     108553 non-null  object 
 1   Code                       103141 non-null  object 
 2   Year                       108553 non-null  object 
 3   Schizophrenia (%)          25875 non-null   object 
 4   Bipolar disorder (%)       19406 non-null   object 
 5   Eating disorders (%)       100236 non-null  object 
 6   Anxiety disorders (%)      6468 non-null    float64
 7   Drug use disorders (%)     6468 non-null    float64
 8   Depression (%)             6468 non-null    float64
 9   Alcohol use disorders (%)  6468 non-null    float64
dtypes: float64(4), object(6)
memory usage: 8.3+ MB
```

In [72]:

```
table_1_data = raw_data.iloc[:6468].copy()
```

In [73]:

```
table_1_data.head()
```

Out[73]:

	Entity	Code	Year	Schizophrenia (%)	Bipolar disorder (%)	Eating disorders (%)	Anxiety disorders (%)	Drug use disorders (%)	Depression (%)	Alcohol use disorders (%)
0	Afghanistan	AFG	1990	0.16056	0.697779	0.101855	4.828830	1.677082	4.071831	0.672404
1	Afghanistan	AFG	1991	0.160312	0.697961	0.099313	4.829740	1.684746	4.079531	0.671768
2	Afghanistan	AFG	1992	0.160135	0.698107	0.096692	4.831108	1.694334	4.088358	0.670644
3	Afghanistan	AFG	1993	0.160037	0.698257	0.094336	4.830864	1.705320	4.096190	0.669738
4	Afghanistan	AFG	1994	0.160022	0.698469	0.092439	4.829423	1.716069	4.099582	0.669260

In [74]:

```
table_1_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6468 entries, 0 to 6467
Data columns (total 10 columns):
 #   Column                     Non-Null Count  Dtype  
---  ------                     --------------  -----  
 0   Entity                     6468 non-null   object 
 1   Code                       5488 non-null   object 
 2   Year                       6468 non-null   object 
 3   Schizophrenia (%)          6468 non-null   object 
 4   Bipolar disorder (%)       6468 non-null   object 
 5   Eating disorders (%)       6468 non-null   object 
 6   Anxiety disorders (%)      6468 non-null   float64
 7   Drug use disorders (%)     6468 non-null   float64
 8   Depression (%)             6468 non-null   float64
 9   Alcohol use disorders (%)  6468 non-null   float64
dtypes: float64(4), object(6)
memory usage: 505.4+ KB
```

In [77]:

```
table_1_data.dropna(axis=0, inplace = True)
```

In [78]:

```
table_1_data.iloc[:, 2:] = table_1_data.iloc[:, 2:].apply(pd.to_numeric)
```

In [79]:

```
table_1_data
```

Out[79]:

	Entity	Code	Year	Schizophrenia (%)	Bipolar disorder (%)	Eating disorders (%)	Anxiety disorders (%)	Drug use disorders (%)	Depression (%)	Alcohol use disorders (%)
0	Afghanistan	AFG	1990	0.160560	0.697779	0.101855	4.828830	1.677082	4.071831	0.672404
1	Afghanistan	AFG	1991	0.160312	0.697961	0.099313	4.829740	1.684746	4.079531	0.671768
2	Afghanistan	AFG	1992	0.160135	0.698107	0.096692	4.831108	1.694334	4.088358	0.670644
3	Afghanistan	AFG	1993	0.160037	0.698257	0.094336	4.830864	1.705320	4.096190	0.669738
4	Afghanistan	AFG	1994	0.160022	0.698469	0.092439	4.829423	1.716069	4.099582	0.669260
...	...	...	...	...	...	...	...	...	...	...
6463	Zimbabwe	ZWE	2013	0.155670	0.607993	0.117248	3.090168	0.766280	3.128192	1.515641
6464	Zimbabwe	ZWE	2014	0.155993	0.608610	0.118073	3.093964	0.768914	3.140290	1.515470
6465	Zimbabwe	ZWE	2015	0.156465	0.609363	0.119470	3.098687	0.771802	3.155710	1.514751
6466	Zimbabwe	ZWE	2016	0.157111	0.610234	0.121456	3.104294	0.772275	3.174134	1.513269
6467	Zimbabwe	ZWE	2017	0.157963	0.611242	0.124443	3.110926	0.772648	3.192789	1.510943

5488 rows × 10 columns

In [80]:

```
table_1_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 5488 entries, 0 to 6467
Data columns (total 10 columns):
 #   Column                     Non-Null Count  Dtype
---  ------                     --------------  -----
 0   Entity                     5488 non-null   object
 1   Code                       5488 non-null   object
 2   Year                       5488 non-null   int64
 3   Schizophrenia (%)          5488 non-null   float64
 4   Bipolar disorder (%)       5488 non-null   float64
 5   Eating disorders (%)       5488 non-null   float64
 6   Anxiety disorders (%)      5488 non-null   float64
 7   Drug use disorders (%)     5488 non-null   float64
 8   Depression (%)             5488 non-null   float64
 9   Alcohol use disorders (%)  5488 non-null   float64
dtypes: float64(7), int64(1), object(2)
memory usage: 471.6+ KB
```

In [81]:

```
## Process each table in the raw table in a separate data frame

## Table 1
mental_health_disorder_share = table_1_data.iloc[:6468].copy()
mental_health_disorder_share.columns = ['country', 'country_code', 'year','schizophrenia', 'bipolar','eating', 'anxiety','drug_use', 'depr
mental_health_disorder_share.iloc[:, 2:] = mental_health_disorder_share.iloc[:, 2:].apply(pd.to_numeric)

## Table 2
disorder_per_sex = table_1_data.iloc[6469:54276, :6].copy()
disorder_per_sex.columns = ['country', 'country_code', 'year','males_share', 'females_share','population']
disorder_per_sex.iloc[:, 3:] = disorder_per_sex.iloc[:, 3:].apply(pd.to_numeric)

## Table 3
suicide_depression = table_1_data.iloc[54277:102084, :6].copy()
suicide_depression.columns = ['country', 'country_code', 'year','suicide', 'depression','population']
suicide_depression.iloc[:, 3:] = suicide_depression.iloc[:, 3:].apply(pd.to_numeric)

## Table 4
depression_prevalence = table_1_data.iloc[102085:, :4].copy()
depression_prevalence.columns = ['country', 'country_code', 'year','prevalence']
depression_prevalence.iloc[:, 3:] = depression_prevalence.iloc[:, 3:].apply(pd.to_numeric)

## Helper Table with Global Popluation for aggregation of Shares
population_per_year = (disorder_per_sex
                           .dropna()
                           .loc[disorder_per_sex.country_code != 'OWID_WRL',['country_code', 'year', 'population']]
                           .copy()
                       )
population_per_year['year'] = pd.to_numeric(population_per_year['year'])
population_per_year['population'] = pd.to_numeric(population_per_year['population'])
```

In [ ]:

```
```

In [82]:

```
# Load global Geometries from naturalearth
world = load_gpd_world()
world = world.rename(columns={'iso_a3':'country_code'})

mental_health_disorder_share = world[['country_code', 'continent', 'geometry']].merge(mental_health_disorder_share)
```

```
C:\Users\Elena Vallikappan\AppData\Local\Temp\ipykernel_22388\349703681.py:2: FutureWarning:

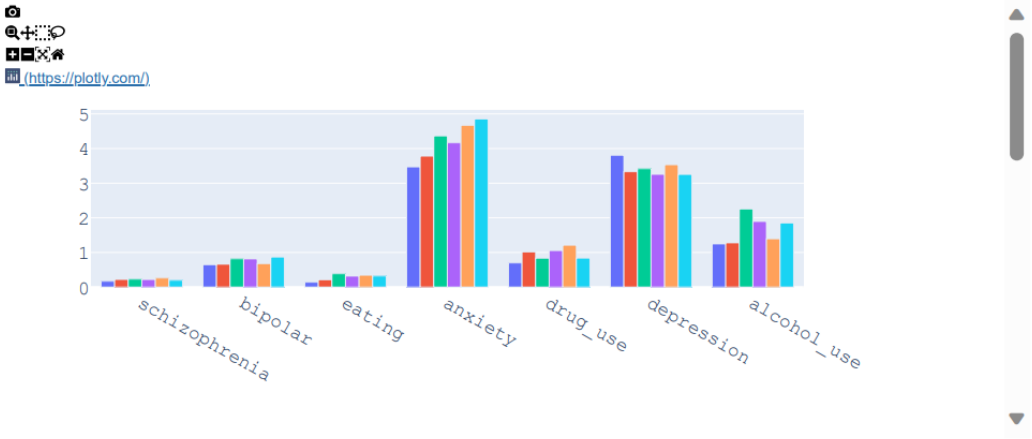
The geopandas.dataset module is deprecated and will be removed in GeoPandas 1.0. You can get the original 'naturalearth_low
res' data from https://www.naturalearthdata.com/downloads/110m-cultural-vectors/. (https://www.naturalearthdata.com/downloa
ds/110m-cultural-vectors/)
```

In [83]:

```
plot_df = (mental_health_disorder_share
               .loc[mental_health_disorder_share.year == 2017]
               .groupby('continent')
               .mean()
               .reset_index()
               .drop(columns=['year'])
               .melt(id_vars='continent')
          )

fig = px.bar(plot_df, x="variable", y="value",
             color="continent", barmode="group",
             title="Mental Health Disorder per Continent in 2017",
             height=400)

fig.update_layout(
    yaxis_title="Share in %",
    xaxis_title="",
    font_family="Courier New",
    font_size=18,
    legend_title="",
)
fig.show()
```

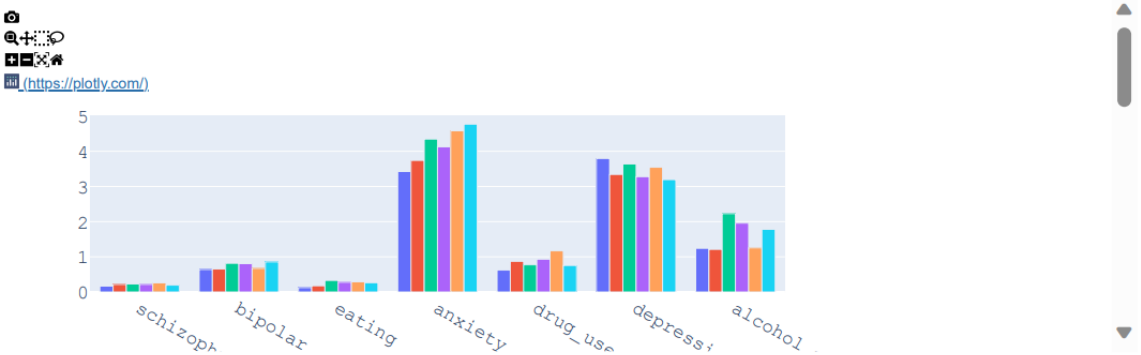


In [84]:

```
plot_df = (mental_health_disorder_share
               .loc[mental_health_disorder_share.year == 1990]
               .groupby('continent')
               .mean()
               .reset_index()
               .drop(columns=['year'])
               .melt(id_vars='continent')
          )

fig = px.bar(plot_df, x="variable", y="value",
             color="continent", barmode="group",
             title="Mental Health Disorder per Continent in 1990",
             height=400)

fig.update_layout(
    yaxis_title="Share in %",
    xaxis_title="",
    font_family="Courier New",
    font_size=18,
    legend_title="",
)
fig.show()
```



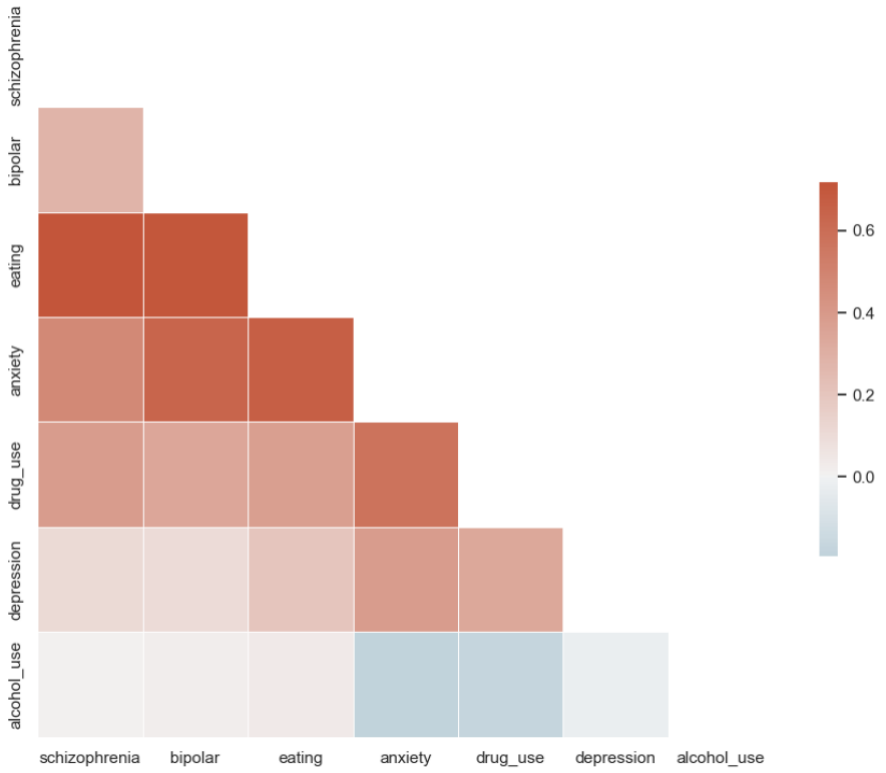
In [85]:

```
sns.set_theme(style="white")
disorder_correlation = mental_health_disorder_share.loc[mental_health_disorder_share.year == 2017, ['schizophrenia','bipolar','eating','a
# Generate a mask for the upper triangle
mask = np.triu(np.ones_like(disorder_correlation, dtype=bool))
highest_correlation = disorder_correlation[disorder_correlation != 1].max().max()

# Set up the matplotlib figure
f, ax = plt.subplots(figsize=(11, 9))

# Generate a custom diverging colormap
cmap = sns.diverging_palette(230, 20, as_cmap=True)

# Draw the heatmap with the mask and correct aspect ratio
_ = sns.heatmap(disorder_correlation, mask=mask, cmap=cmap, vmax=highest_correlation, center=0,
            square=True, linewidths=.5, cbar_kws={"shrink": .5})
```



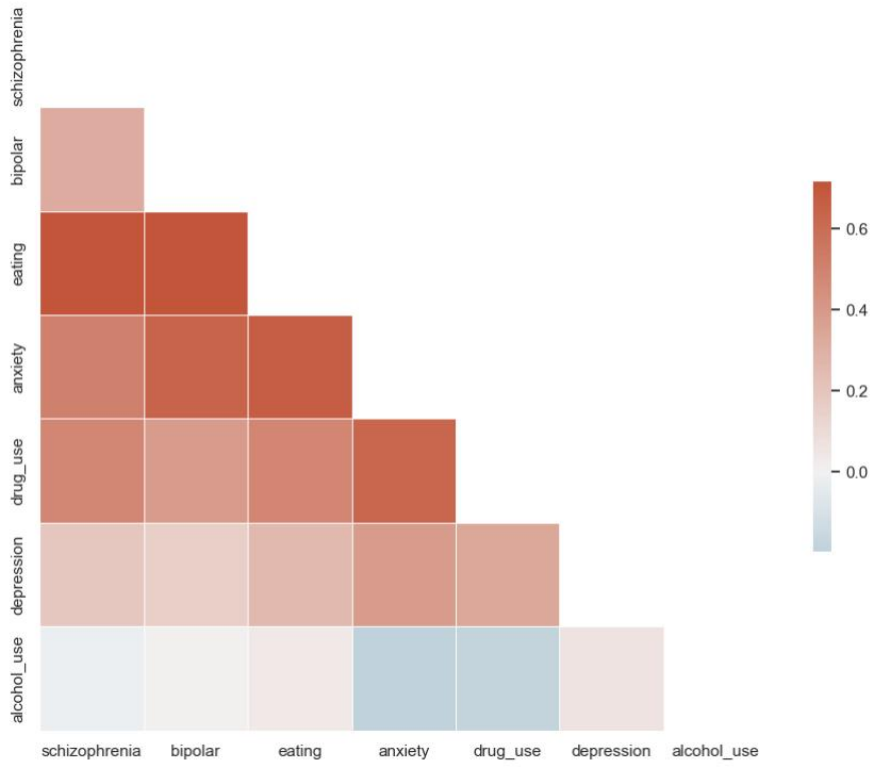
In [86]:

```
sns.set_theme(style="white")
disorder_correlation = mental_health_disorder_share.loc[mental_health_disorder_share.year == 1990, ['schizophrenia','bipolar','eating','a
# Generate a mask for the upper triangle
mask = np.triu(np.ones_like(disorder_correlation, dtype=bool))
highest_correlation = disorder_correlation[disorder_correlation != 1].max().max()

# Set up the matplotlib figure
f, ax = plt.subplots(figsize=(11, 9))

# Generate a custom diverging colormap
cmap = sns.diverging_palette(230, 20, as_cmap=True)

# Draw the heatmap with the mask and correct aspect ratio
_ = sns.heatmap(disorder_correlation, mask=mask, cmap=cmap, vmax=highest_correlation, center=0,
            square=True, linewidths=.5, cbar_kws={"shrink": .5})
```



In [87]:

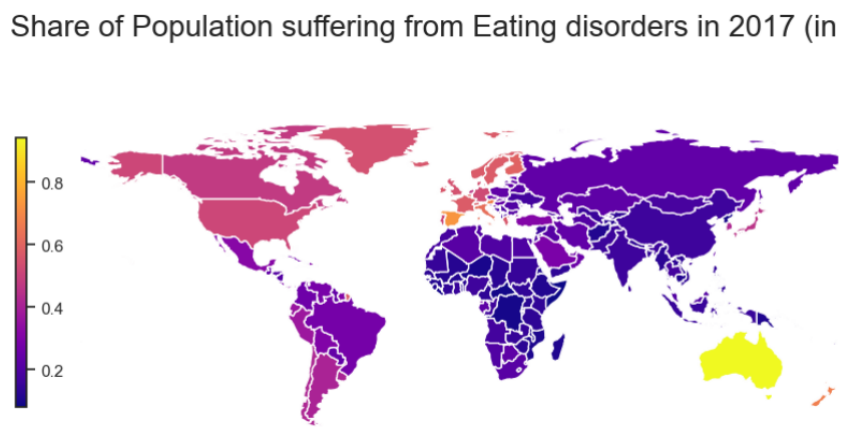
```
plot_df = mental_health_disorder_share.loc[mental_health_disorder_share.year == 2017]

cmap='plasma'
fig, ax = plt.subplots(figsize=(10, 5))

measure_column = 'eating'

plot_df.plot(edgecolor='white',
                            linewidth=1,
                            column=measure_column,
                            cmap=cmap,
                            ax=ax)

sm = plt.cm.ScalarMappable(norm=plt.Normalize(vmin=plot_df[measure_column].min(), vmax=plot_df[measure_column].max()), cmap=cmap)
cbaxes = fig.add_axes([0.1, 0.25, 0.01, 0.5])
cbar = fig.colorbar(sm, cax=cbaxes)
fig.suptitle("Share of Population suffering from Eating disorders in 2017 (in %)", fontsize=20)
ax.axis('off')
plt.show()
```



In [88]:

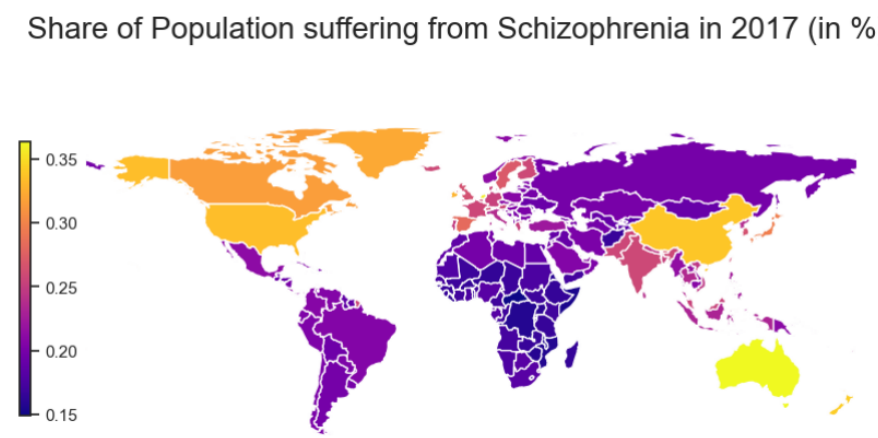
```
plot_df = mental_health_disorder_share.loc[mental_health_disorder_share.year == 2017]

cmap='plasma'
fig, ax = plt.subplots(figsize=(10, 5))

measure_column = 'schizophrenia'

plot_df.plot(edgecolor='white',
                            linewidth=1,
                            column=measure_column,
                            cmap=cmap,
                            ax=ax)

sm = plt.cm.ScalarMappable(norm=plt.Normalize(vmin=plot_df[measure_column].min(), vmax=plot_df[measure_column].max()), cmap=cmap)
cbaxes = fig.add_axes([0.1, 0.25, 0.01, 0.5])
cbar = fig.colorbar(sm, cax=cbaxes)
fig.suptitle("Share of Population suffering from Schizophrenia in 2017 (in %)", fontsize=20)
ax.axis('off')
plt.show()
```



In [89]:

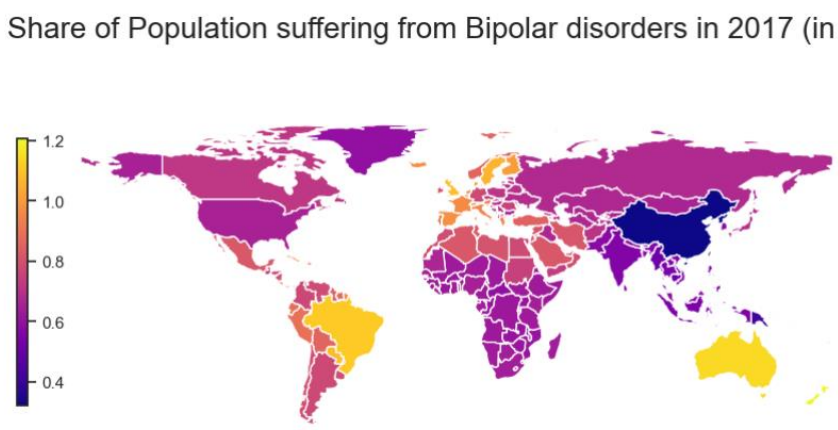
```
plot_df = mental_health_disorder_share.loc[mental_health_disorder_share.year == 2017]

cmap='plasma'
fig, ax = plt.subplots(figsize=(10, 5))

measure_column = 'bipolar'

plot_df.plot(edgecolor='white',
                            linewidth=1,
                            column=measure_column,
                            cmap=cmap,
                            ax=ax)

sm = plt.cm.ScalarMappable(norm=plt.Normalize(vmin=plot_df[measure_column].min(), vmax=plot_df[measure_column].max()), cmap=cmap)
cbaxes = fig.add_axes([0.1, 0.25, 0.01, 0.5])
cbar = fig.colorbar(sm, cax=cbaxes)
fig.suptitle("Share of Population suffering from Bipolar disorders in 2017 (in %)", fontsize=20)
ax.axis('off')
plt.show()
```



In [90]:

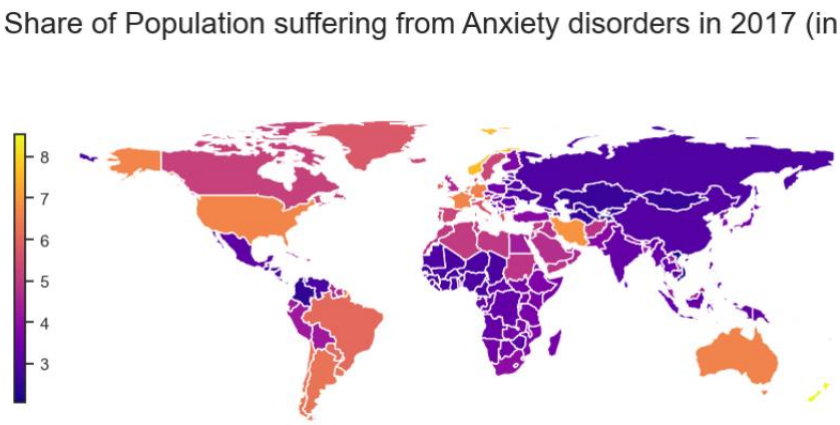
```
plot_df = mental_health_disorder_share.loc[mental_health_disorder_share.year == 2017]

cmap='plasma'
fig, ax = plt.subplots(figsize=(10, 5))

measure_column = 'anxiety'

plot_df.plot(edgecolor='white',
                            linewidth=1,
                            column=measure_column,
                            cmap=cmap,
                            ax=ax)

sm = plt.cm.ScalarMappable(norm=plt.Normalize(vmin=plot_df[measure_column].min(), vmax=plot_df[measure_column].max()), cmap=cmap)
cbaxes = fig.add_axes([0.1, 0.25, 0.01, 0.5])
cbar = fig.colorbar(sm, cax=cbaxes)
fig.suptitle("Share of Population suffering from Anxiety disorders in 2017 (in %)", fontsize=20)
ax.axis('off')
plt.show()
```



In [91]:

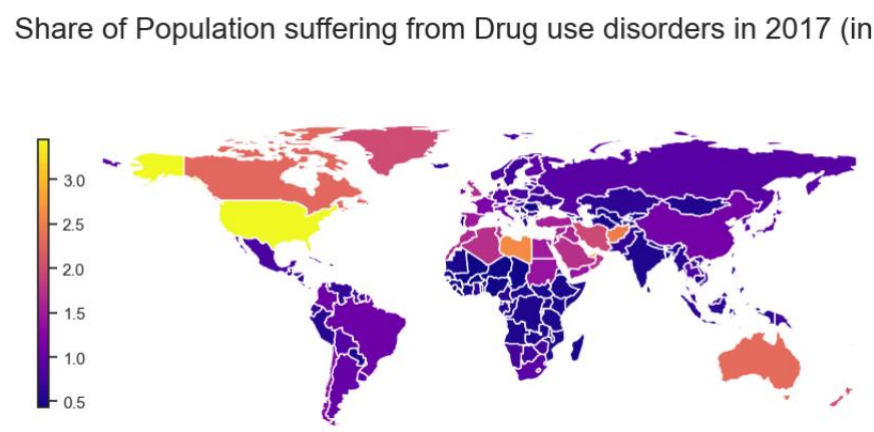
```
plot_df = mental_health_disorder_share.loc[mental_health_disorder_share.year == 2017]

cmap='plasma'
fig, ax = plt.subplots(figsize=(10, 5))

measure_column = 'drug_use'

plot_df.plot(edgecolor='white',
                            linewidth=1,
                            column=measure_column,
                            cmap=cmap,
                            ax=ax)

sm = plt.cm.ScalarMappable(norm=plt.Normalize(vmin=plot_df[measure_column].min(), vmax=plot_df[measure_column].max()), cmap=cmap)
cbaxes = fig.add_axes([0.1, 0.25, 0.01, 0.5])
cbar = fig.colorbar(sm, cax=cbaxes)
fig.suptitle("Share of Population suffering from Drug use disorders in 2017 (in %)", fontsize=20)
ax.axis('off')
plt.show()
```



In [92]:

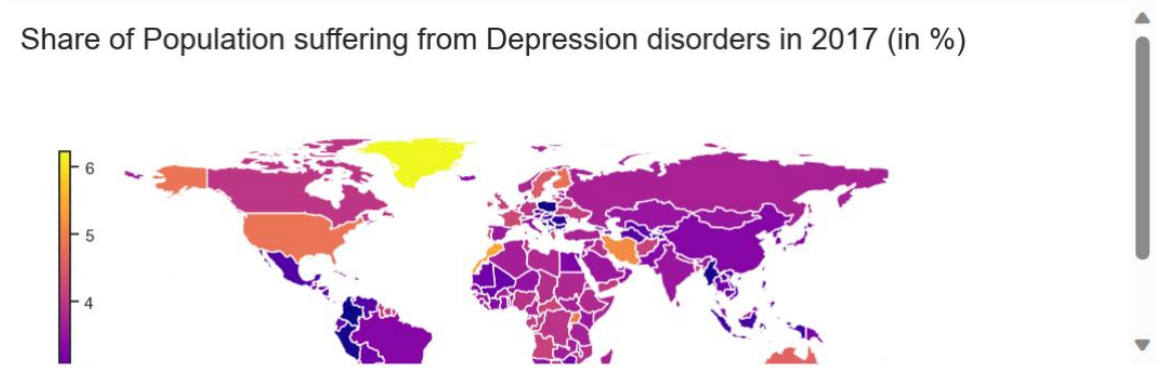
```
plot_df = mental_health_disorder_share.loc[mental_health_disorder_share.year == 2017]

cmap='plasma'
fig, ax = plt.subplots(figsize=(10, 5))

measure_column = 'depression'

plot_df.plot(edgecolor='white',
                            linewidth=1,
                            column=measure_column,
                            cmap=cmap,
                            ax=ax)

sm = plt.cm.ScalarMappable(norm=plt.Normalize(vmin=plot_df[measure_column].min(), vmax=plot_df[measure_column].max()), cmap=cmap)
cbaxes = fig.add_axes([0.1, 0.25, 0.01, 0.5])
cbar = fig.colorbar(sm, cax=cbaxes)
fig.suptitle("Share of Population suffering from Depression disorders in 2017 (in %)", fontsize=20)
ax.axis('off')
plt.show()
```



In [93]:

```
plot_df = mental_health_disorder_share.loc[mental_health_disorder_share.year == 2017]

cmap='plasma'
fig, ax = plt.subplots(figsize=(10, 5))

measure_column = 'alcohol_use'

plot_df.plot(edgecolor='white',
                            linewidth=1,
                            column=measure_column,
                            cmap=cmap,
                            ax=ax)

sm = plt.cm.ScalarMappable(norm=plt.Normalize(vmin=plot_df[measure_column].min(), vmax=plot_df[measure_column].max()), cmap=cmap)
cbaxes = fig.add_axes([0.1, 0.25, 0.01, 0.5])
cbar = fig.colorbar(sm, cax=cbaxes)
fig.suptitle("Share of Population suffering from Alcohol use disorders in 2017 (in %)", fontsize=20)
ax.axis('off')
plt.show()
```



In [94]:

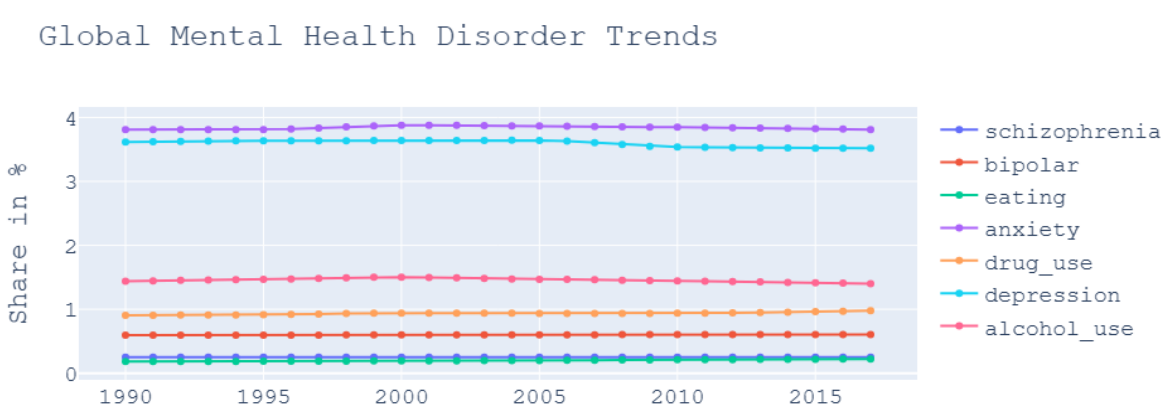
```
disorders_yearly = mental_health_disorder_share.merge(population_per_year, on =['country_code','year'])
# Transform into total population counts
disorders_yearly.iloc[:, 5:-1] = disorders_yearly.iloc[:, 5:-1].apply(lambda x: (x/100) * disorders_yearly.iloc[:,-1])

disorders_yearly = (disorders_yearly
     .groupby('year')
     .sum()
     .reset_index()
)
# Calculate the Share again
disorders_yearly.iloc[:, 1:-1] = disorders_yearly.iloc[:, 1:-1].apply(lambda x : (x / disorders_yearly.iloc[:,-1]) * 100)

disorders_yearly = (disorders_yearly
                        .drop(columns='population')
                        .melt(id_vars='year')
                        .rename(columns={'variable':'disease', 'value':'share'})
                )

fig = px.line(disorders_yearly, x='year', y='share', color='disease', markers=True,
             title="Global Mental Health Disorder Trends",
             height=400)

fig.update_layout(
    yaxis_title="Share in %",
    xaxis_title="",
    font_family="Courier New",
    font_size=18,
    legend_title="",
)
fig.show()
```



In [59]:

```
disorders_regional_change = mental_health_disorder_share.copy()
disorders_regional_change['total_disorders_share'] = disorders_regional_change.iloc[:, 5:].sum(axis=1)
disorders_regional_change = disorders_regional_change[['country', 'country_code', 'year','total_disorders_share']]
disorders_regional_change = disorders_regional_change.sort_values(['country','year'])

disorders_regional_change = disorders_regional_change.groupby(['country', 'country_code']).agg({'total_disorders_share':['first','last']})
disorders_regional_change.columns =['country','country_code','first','last']
disorders_regional_change['change'] = disorders_regional_change.iloc[:, -1] - disorders_regional_change.iloc[:, -2]

disorders_regional_change = world[['country_code', 'continent', 'geometry']].merge(disorders_regional_change)
```

In [62]:

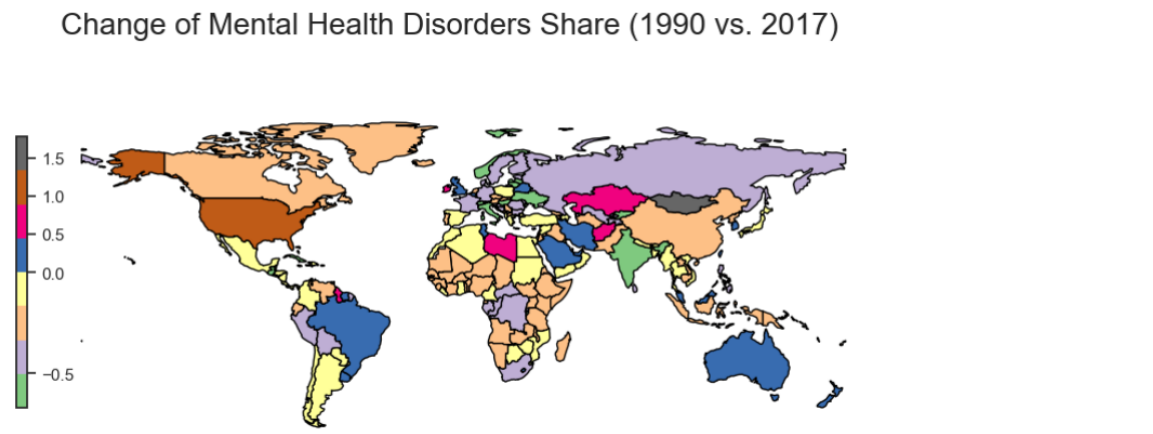
```
from matplotlib.colors import TwoSlopeNorm

cmap='Accent'
fig, ax = plt.subplots(figsize=(10, 5))

disorders_regional_change.plot(edgecolor='black',
                               linewidth=1,
                               column='change',
                               cmap=cmap,
                               ax=ax)

sm = plt.cm.ScalarMappable(norm=TwoSlopeNorm(0, vmin=disorders_regional_change['change'].min(), vmax=disorders_regional_change['change'].

cbaxes = fig.add_axes([0.1, 0.25, 0.01, 0.5])
cbar = fig.colorbar(sm, cax=cbaxes)
fig.suptitle("Change of Mental Health Disorders Share (1990 vs. 2017)", fontsize=20)
ax.axis('off')
plt.show()
```



In [96]:

```
raw_data.dtypes
```

Out[96]:

```
Entity                         object
Code                           object
Year                           object
Schizophrenia (%)              object
Bipolar disorder (%)           object
Eating disorders (%)           object
Anxiety disorders (%)         float64
Drug use disorders (%)        float64
Depression (%)                float64
Alcohol use disorders (%)     float64
dtype: object
```

In [97]:

```
raw_data.duplicated().sum()
```

Out[97]:

```
9718
```

In [98]:

```
duplicated_data = raw_data[raw_data.duplicated(keep='first')==True]
duplicated_data.count()
```

Out[98]:

```
Entity                        9718
Code                          9474
Year                          9718
Schizophrenia (%)                0
Bipolar disorder (%)             0
Eating disorders (%)          9718
Anxiety disorders (%)            0
Drug use disorders (%)           0
Depression (%)                   0
Alcohol use disorders (%)        0
dtype: int64
```

In [100]:

```
# drop the duplicated rows, because in this analysis we don't need the duplicated datas.
raw_data.drop_duplicates(keep='first', inplace=True)
```

In [101]:

```
# check null values in df
raw_data[raw_data.isna().any(axis=1)==True]      # here the null value is also present in 'Code' column, but this column does not affect m
```

Out[101]:

	Entity	Code	Year	Schizophrenia (%)	Bipolar disorder (%)	Eating disorders (%)	Anxiety disorders (%)	Drug use disorders (%)	Depression (%)	Alcohol use disorders (%)
112	Andean Latin America	NaN	1990	0.195471	0.870564	0.28782	4.217677	0.608921	2.621781	1.748912
113	Andean Latin America	NaN	1991	0.195484	0.870876	0.289362	4.220261	0.610489	2.631717	1.743910
114	Andean Latin America	NaN	1992	0.195497	0.871174	0.290563	4.222587	0.611990	2.639932	1.738703
115	Andean Latin America	NaN	1993	0.195523	0.871482	0.291833	4.224721	0.613995	2.646096	1.733697
116	Andean Latin America	NaN	1994	0.195569	0.871823	0.293111	4.226690	0.616448	2.651022	1.729266
...	...	...	...	...	...	...	...	...	...	...
108548	Zimbabwe	ZWE	2013	303564.6036	NaN	NaN	NaN	NaN	NaN	NaN
108549	Zimbabwe	ZWE	2014	311665.7693	NaN	NaN	NaN	NaN	NaN	NaN
108550	Zimbabwe	ZWE	2015	320638.5072	NaN	NaN	NaN	NaN	NaN	NaN
108551	Zimbabwe	ZWE	2016	330437.3538	NaN	NaN	NaN	NaN	NaN	NaN
108552	Zimbabwe	ZWE	2017	340583.1574	NaN	NaN	NaN	NaN	NaN	NaN

93347 rows × 10 columns

In [102]:

```
# create a new dataframe named 'req_df' to clean dataframe more effectively.
req_headers = ['Entity', 'Year', 'Schizophrenia (%)', 'Bipolar disorder (%)',
       'Eating disorders (%)', 'Anxiety disorders (%)',
       'Drug use disorders (%)', 'Depression (%)',
       'Alcohol use disorders (%)']
req_df = raw_data[req_headers]
req_df.head()
```

Out[102]:

	Entity	Year	Schizophrenia (%)	Bipolar disorder (%)	Eating disorders (%)	Anxiety disorders (%)	Drug use disorders (%)	Depression (%)	Alcohol use disorders (%)
0	Afghanistan	1990	0.16056	0.697779	0.101855	4.828830	1.677082	4.071831	0.672404
1	Afghanistan	1991	0.160312	0.697961	0.099313	4.829740	1.684746	4.079531	0.671768
2	Afghanistan	1992	0.160135	0.698107	0.096692	4.831108	1.694334	4.088358	0.670644
3	Afghanistan	1993	0.160037	0.698257	0.094336	4.830864	1.705320	4.096190	0.669738
4	Afghanistan	1994	0.160022	0.698469	0.092439	4.829423	1.716069	4.099582	0.669260

In [103]:

```
# cleaning the 'req_df' more
req_df.isna().any(axis=0).sum()
```

Out[103]:

```
7
```

In [104]:

```
# drop null values from dataframe
req_df.dropna(axis=0, inplace=True)
```

```
C:\Users\Elena Vallikappan\AppData\Local\Temp\ipykernel_22388\3625022547.py:2: SettingWithCopyWarning:


A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy (https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy)
```

In [105]:

```
# check if any null values is still remaining
req_df.isna().sum()
```

Out[105]:

```
Entity                        0
Year                          0
Schizophrenia (%)             0
Bipolar disorder (%)          0
Eating disorders (%)          0
Anxiety disorders (%)         0
Drug use disorders (%)        0
Depression (%)                0
Alcohol use disorders (%)     0
dtype: int64
```

In [108]:

```
# data_check stats

data_check_table = pd.DataFrame({
    'Unique':raw_data.nunique(),
    'null':raw_data.isna().sum(),
    'null percent':(raw_data.isna().sum())/len(raw_data)*100,
    'Type':raw_data.dtypes.values
})

data_check_table
```

Out[108]:

	Unique	null	null percent	Type
Entity	276	0	0.000000	object
Code	236	5168	5.228917	object
Year	259	0	0.000000	object
Schizophrenia (%)	25661	72960	73.820003	object
Bipolar disorder (%)	19358	79429	80.365255	object
Eating disorders (%)	72657	8317	8.415035	object
Anxiety disorders (%)	6457	92367	93.455760	float64
Drug use disorders (%)	6435	92367	93.455760	float64
Depression (%)	6460	92367	93.455760	float64
Alcohol use disorders (%)	6459	92367	93.455760	float64

In [109]:

```
# converting the datatype of object as per necessity
def convert_str_to_float(x):
    if type(x)!=float:
        if x[0] in ['0','1','2','3','4','5','6','7','8','9']:
            return float(x)
        return float(0)
    return x

for i in req_df.columns[2:]:
    req_df[i]=req_df[i].apply(convert_str_to_float)
```

```
C:\Users\Elena Vallikappan\AppData\Local\Temp\ipykernel_22388\3163692857.py:10: SettingWithCopyWarning:


A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-vie
w-versus-a-copy (https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy)
```

In [110]:

```
for i in req_df.columns[2:]:
    print(i)
```

```
Schizophrenia (%)
Bipolar disorder (%)
Eating disorders (%)
Anxiety disorders (%)
Drug use disorders (%)
Depression (%)
Alcohol use disorders (%)
```

In [111]:

```
# Exploratory Data Analysis
"""
    EDA on the mean of every issue with respect to Entity & Year
"""

# grouped datas
nation_group_issue_avg = req_df.groupby('Entity', as_index = False).agg({'Schizophrenia (%)':'mean', 'Bipolar disorder (%)':'mean','Eating
```

In [112]:

```
for i in nation_group_issue_avg.columns[1:]:
    print(i)
```

```
Schizophrenia (%)
Bipolar disorder (%)
Eating disorders (%)
Anxiety disorders (%)
Drug use disorders (%)
Depression (%)
Alcohol use disorders (%)
```

In [113]:

```
import matplotlib.pyplot as plt
import random


colors = ['red', 'orange', 'purple']

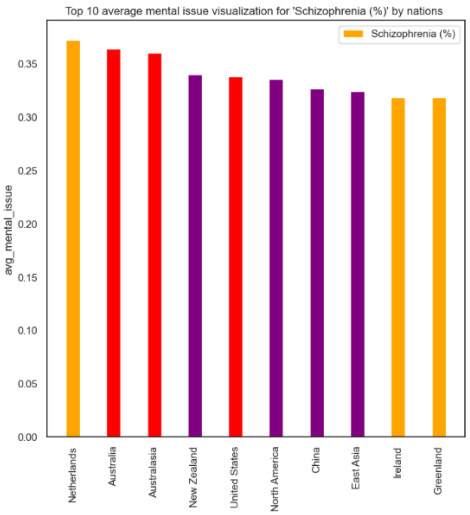

for i in nation_group_issue_avg.columns[1:]:
    top_10_nation_group_issue_avg = nation_group_issue_avg.sort_values(by=i, ascending=False).head(10)

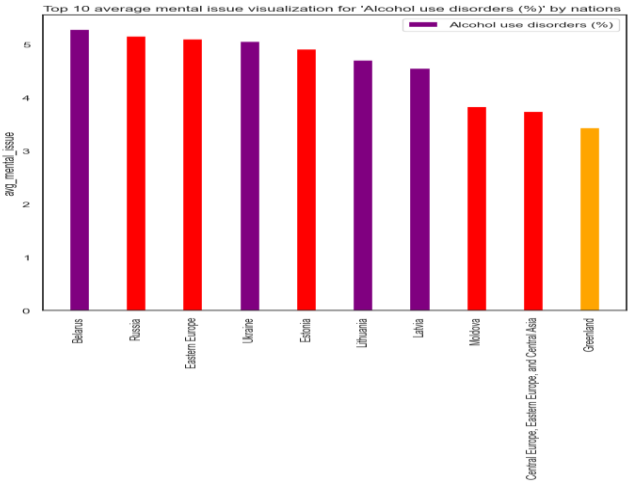
    # generates the random colors for bars in the figure.
    bar_colors = []
    for j in range(20):
        bar_colors.append(random.choice(colors))

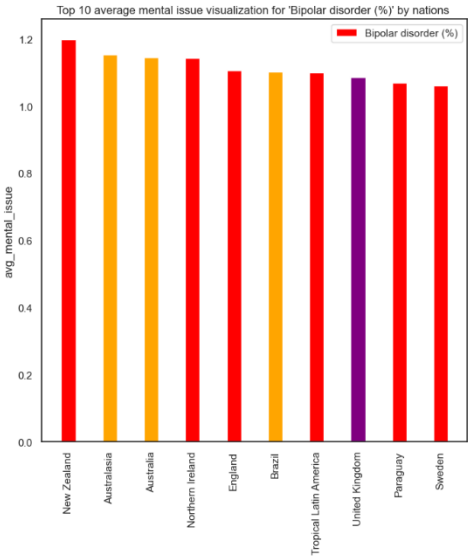
    fig, ax = plt.subplots(figsize=(8,8))
    ax.bar(top_10_nation_group_issue_avg['Entity'],top_10_nation_group_issue_avg[i] , width=0.35, label=i, color=bar_colors)

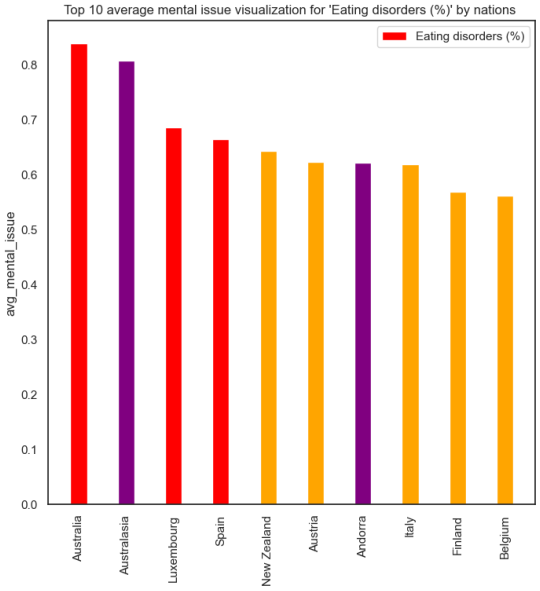
    ax.set_ylabel('avg_mental_issue')
    ax.set_title(f"Top 10 average mental issue visualization for '{i}' by nations ")
    plt.xticks(rotation=90)
    ax.legend()

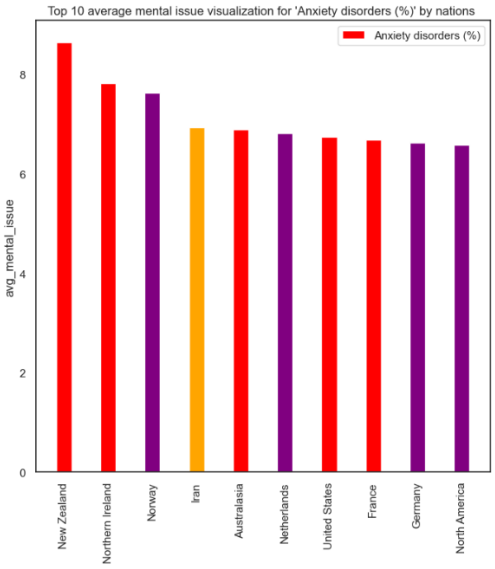
    plt.show()
```

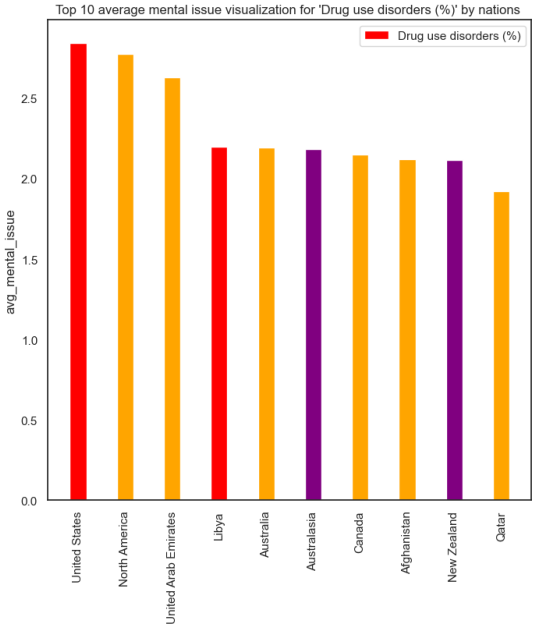


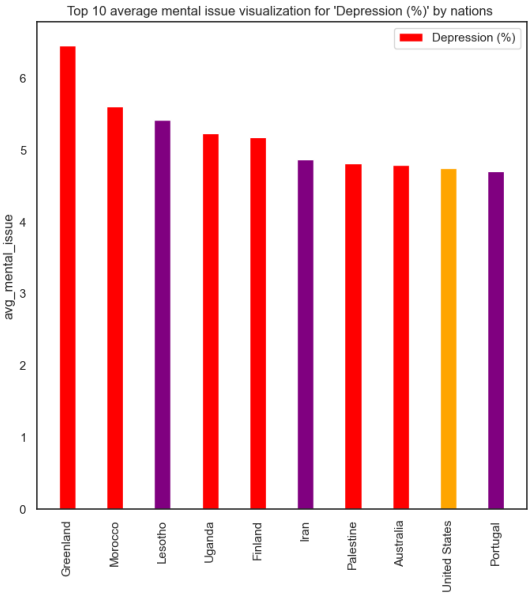





Top 10 average mental issue visualization for 'Eating disorders (%)' by nations
Eating disorders (%)
avg_mental_issue
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Australia
Australasia
Luxembourg
Spain
New Zealand
Austria
Andorra
Italy
Finland
Belgium


Top 10 average mental issue visualization for 'Anxiety disorders (%)' by nations
Anxiety disorders (%)
avg_mental_issue
0
2
4
6
8
New Zealand
Northern Ireland
Norway
Iran
Australasia
Netherlands
United States
France
Germany
North America


Top 10 average mental issue visualization for 'Drug use disorders (%)' by nations
Drug use disorders (%)
avg_mental_issue
0.0
0.5
1.0
1.5
2.0
2.5
United States
North America
United Arab Emirates
Libya
Australia
Australasia
Canada
Afghanistan
New Zealand
Qatar


Top 10 average mental issue visualization for 'Depression (%)' by nations
Depression (%)
avg_mental_issue
0
1
2
3
4
5
6
Greenland
Morocco
Lesotho
Uganda
Finland
Iran
Palestine
Australia
United States
Portugal

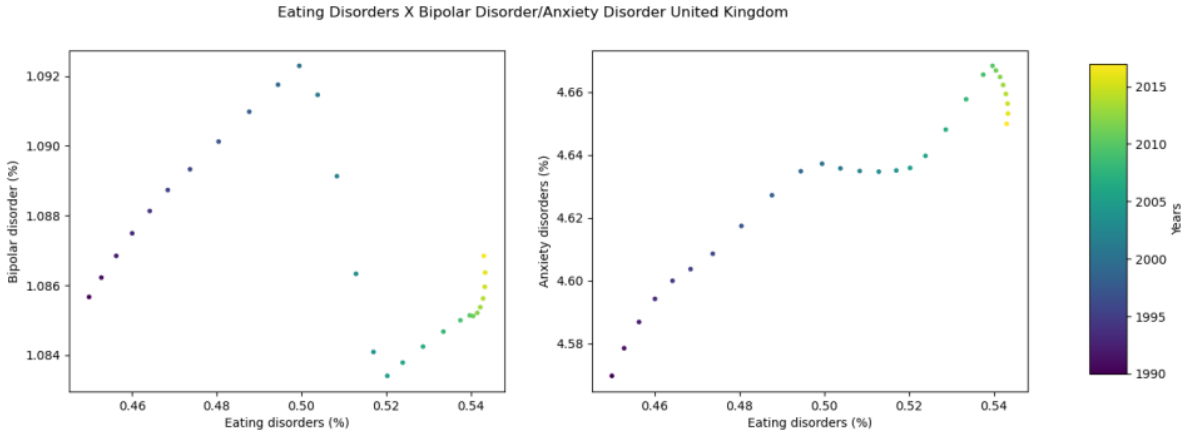
In [36]:

```
fig, ax = plt.subplots(1,2, figsize = (14,5), sharex=True)

brazil = table[table.Entity.isin(['United Kingdom'])]

im = ax[0].scatter(brazil['Eating disorders (%)'], brazil['Bipolar disorder (%)'], c=brazil['Year'], s= 8, alpha= 1);
im = ax[1].scatter(brazil['Eating disorders (%)'], brazil['Anxiety disorders (%)'], c=brazil['Year'], s= 8, alpha= 1);
ax[0].set_xlabel('Eating disorders (%)')
ax[0].set_ylabel('Bipolar disorder (%)')
ax[1].set_xlabel('Eating disorders (%)')
ax[1].set_ylabel('Anxiety disorders (%)')

cbar_ax = fig.add_axes([0.95, 0.15, 0.03, 0.7])
fig.colorbar(im, cax=cbar_ax, label = 'Years')
plt.suptitle('Eating Disorders X Bipolar Disorder/Anxiety Disorder United Kingdom');
```



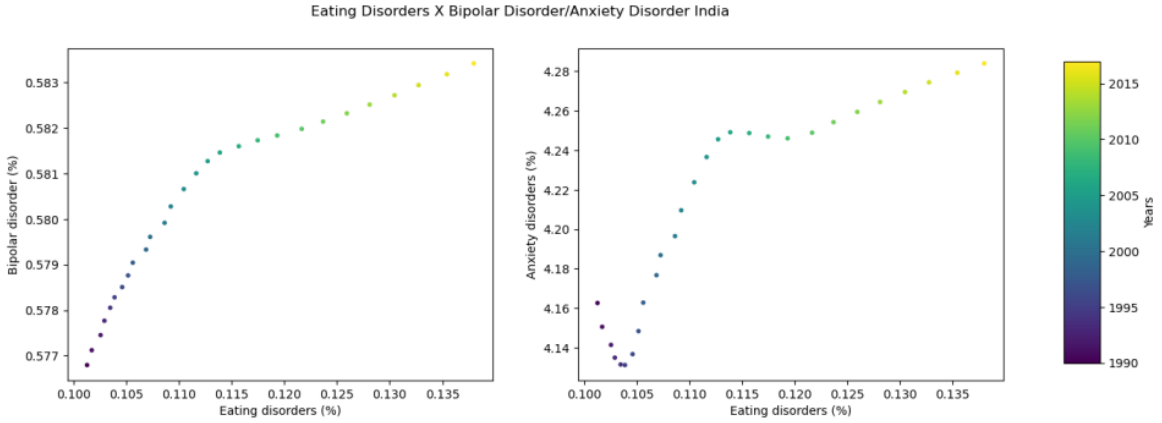
In [44]:

```
fig, ax = plt.subplots(1,2, figsize = (14,5), sharex=True)

brazil = table[table.Entity.isin(['Bangladesh'])]

im = ax[0].scatter(brazil['Eating disorders (%)'], brazil['Bipolar disorder (%)'], c=brazil['Year'], s= 8, alpha= 1);
im = ax[1].scatter(brazil['Eating disorders (%)'], brazil['Anxiety disorders (%)'], c=brazil['Year'], s= 8, alpha= 1);
ax[0].set_xlabel('Eating disorders (%)')
ax[0].set_ylabel('Bipolar disorder (%)')
ax[1].set_xlabel('Eating disorders (%)')
ax[1].set_ylabel('Anxiety disorders (%)')

cbar_ax = fig.add_axes([0.95, 0.15, 0.03, 0.7])
fig.colorbar(im, cax=cbar_ax, label = 'Years')
plt.suptitle('Eating Disorders X Bipolar Disorder/Anxiety Disorder India');
```



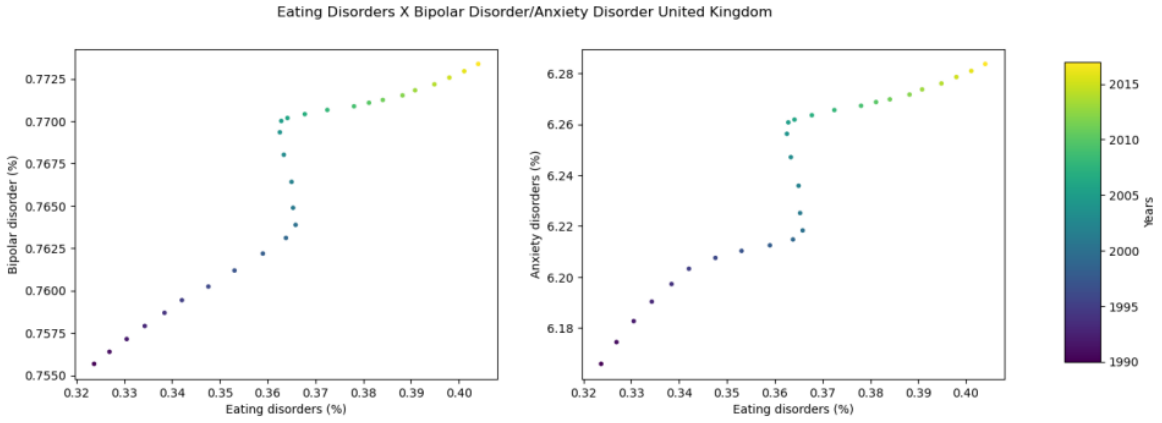
In [38]:

```
fig, ax = plt.subplots(1,2, figsize = (14,5), sharex=True)

brazil = table[table.Entity.isin(['Argentina'])]

im = ax[0].scatter(brazil['Eating disorders (%)'], brazil['Bipolar disorder (%)'], c=brazil['Year'], s= 8, alpha= 1);
im = ax[1].scatter(brazil['Eating disorders (%)'], brazil['Anxiety disorders (%)'], c=brazil['Year'], s= 8, alpha= 1);
ax[0].set_xlabel('Eating disorders (%)')
ax[0].set_ylabel('Bipolar disorder (%)')
ax[1].set_xlabel('Eating disorders (%)')
ax[1].set_ylabel('Anxiety disorders (%)')

cbar_ax = fig.add_axes([0.95, 0.15, 0.03, 0.7])
fig.colorbar(im, cax=cbar_ax, label = 'Years')
plt.suptitle('Eating Disorders X Bipolar Disorder/Anxiety Disorder Argentina');
```



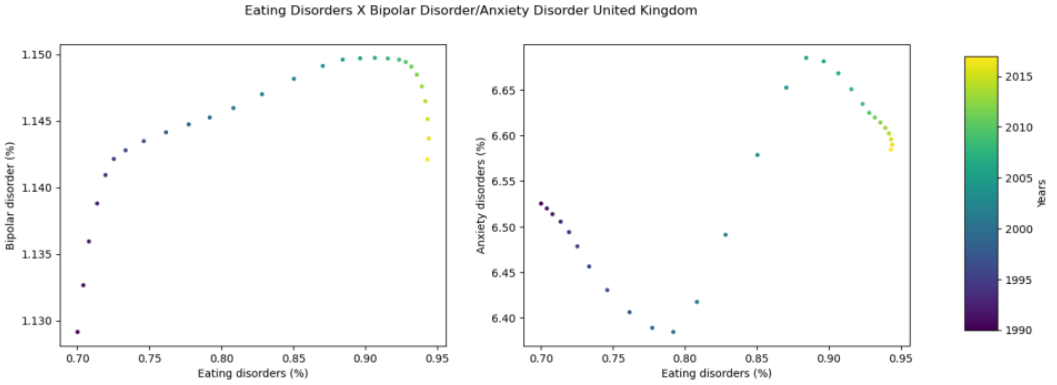
In [39]:

```
fig, ax = plt.subplots(1,2, figsize = (14,5), sharex=True)

brazil = table[table.Entity.isin(['Australia'])]

im = ax[0].scatter(brazil['Eating disorders (%)'], brazil['Bipolar disorder (%)'], c=brazil['Year'], s= 8, alpha= 1);
im = ax[1].scatter(brazil['Eating disorders (%)'], brazil['Anxiety disorders (%)'], c=brazil['Year'], s= 8, alpha= 1);
ax[0].set_xlabel('Eating disorders (%)')
ax[0].set_ylabel('Bipolar disorder (%)')
ax[1].set_xlabel('Eating disorders (%)')
ax[1].set_ylabel('Anxiety disorders (%)')

cbar_ax = fig.add_axes([0.95, 0.15, 0.03, 0.7])
fig.colorbar(im, cax=cbar_ax, label = 'Years')
plt.suptitle('Eating Disorders X Bipolar Disorder/Anxiety Disorder Australia');
```



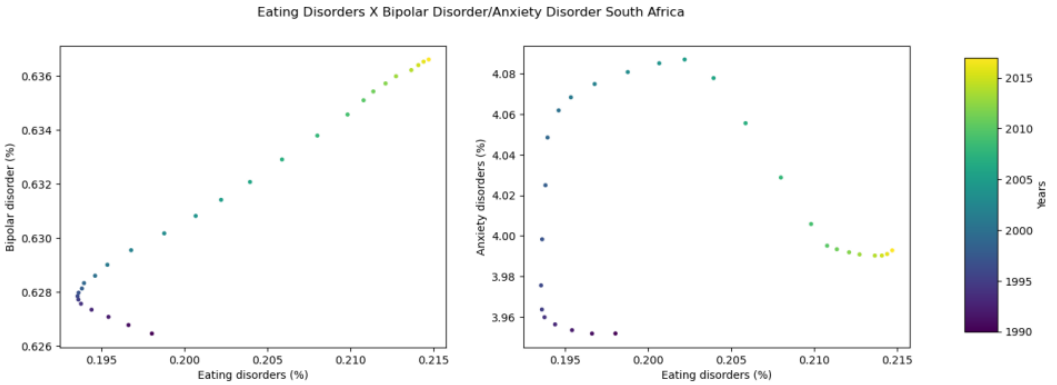
In [43]:

```
fig, ax = plt.subplots(1,2, figsize = (14,5), sharex=True)

brazil = table[table.Entity.isin(['South Africa'])]

im = ax[0].scatter(brazil['Eating disorders (%)'], brazil['Bipolar disorder (%)'], c=brazil['Year'], s= 8, alpha= 1);
im = ax[1].scatter(brazil['Eating disorders (%)'], brazil['Anxiety disorders (%)'], c=brazil['Year'], s= 8, alpha= 1);
ax[0].set_xlabel('Eating disorders (%)')
ax[0].set_ylabel('Bipolar disorder (%)')
ax[1].set_xlabel('Eating disorders (%)')
ax[1].set_ylabel('Anxiety disorders (%)')

cbar_ax = fig.add_axes([0.95, 0.15, 0.03, 0.7])
fig.colorbar(im, cax=cbar_ax, label = 'Years')
plt.suptitle('Eating Disorders X Bipolar Disorder/Anxiety Disorder South Africa');
```



In [47]:

```
Year').mean().plot(figsize=(13,6), title='Disorders x Years (mean over all countries)', ylabel = 'Normalized Values', xlabel = 'Years');
```

